

# Naor-Reingold goes Beyond-the-Birthday-Bound

Avik Chakraborti<sup>1,3</sup>, Bishwajit Chakraborty<sup>2</sup>, Nilanjan Datta<sup>1,3</sup>, Avijit Dutta<sup>1,3</sup>, Ashwin Jha<sup>4</sup>, Sougata Mandal<sup>1,5</sup>, Hrithik Nandi<sup>1,5</sup>, Mridul Nandi<sup>6</sup>,  
and Abishanka Saha<sup>7</sup>

<sup>1</sup> Institute for Advancing Intelligence, TCG CREST, Kolkata, India

<sup>2</sup> Nanyang Technological University, Singapore.

<sup>3</sup> AcSIR, Ghaziabad, India.

<sup>4</sup> Ruhr-University Bochum, Bochum, Germany.

<sup>5</sup> Ramkrishna Mission Vivekananda Educational Research Institute.

<sup>6</sup> Indian Statistical Institute, Kolkata, Kolkata, India.

<sup>7</sup> Eindhoven University, Netherlands.

{avik.chakraborti, nilanjan.datta, sougata.mandal, hrithik.nandi}@tcgcrest.org,  
{bishumathynwa, avirocksdutta13, letterstoashwin, mridulnandi, sahaa1993}@gmail.com,

**Abstract** Construction of efficient and provably-secure (T)PRPs and (fixed/variable input-length) PRFs has been one of the central open problem in modern symmetric-key cryptography. Many Feistel-based constructions has been proposed and analysed to solve this problem. Inspired by some recent works, in this paper, we revisit the problem of constructing provably secure Feistel constructions using permutations as the round functions. More specifically, following the idea of Naor and Reingold, we try to reduce the number of inner permutations used by replacing them with cheap hash functions, without sacrificing optimal security. We affirmatively show that with the use of a suitable hash function along with a four-round Feistel construction, which uses only three independent permutations, one can achieve optimally secure (T)PRPs and PRFs.

## 1 Introduction

Modern cryptography, particularly symmetric-key cryptography, bases its security on the probability that an adversary can successfully break the system. This probabilistic aspect comes from the pseudo-random behavior of the underlying primitive functions. Therefore, designing appropriate pseudo-random functions (PRFs) and pseudo-random permutations (PRPs) has always been a central problem in symmetric-key cryptography. Since PRFs have a stronger security definition, they are generally more difficult to construct than PRPs. As a result, most approaches focus on building PRPs and then deriving PRFs from them.

### 1.1 Building PRPs

A block cipher (BC), also known as a keyed pseudo-random permutation (PRP), is defined over a key space  $\mathcal{K}$  and a message space  $\mathcal{M}$ , and denoted as  $E :$

$\mathcal{K} \times \mathcal{M} \rightarrow \mathcal{M}$ , such that for every  $k \in \mathcal{K}$ , the function  $E_k : \mathcal{M} \rightarrow \mathcal{M}$ , where  $E_k(M) := E(k, M)$ , is a permutation. One of the most well-known block cipher design methods is the Feistel structure, or LR cipher, introduced by Luby and Rackoff in 1986 [13]. They showed that a 3-round LR cipher using random round functions is indistinguishable from a random permutation up to  $\mathcal{O}(2^{n/2})$  queries. Later, Patarin extended this result by proving that a 4-round LR cipher is secure against chosen-ciphertext attacks up to  $\mathcal{O}(2^{n/2})$  queries [18]. In 1999, Naor and Reingold [17] demonstrated that by using a less costly Almost Right-Collision-avoiding Universal (ARCU) hash function, a 2-round LR construction with just 2 random functions can achieve chosen-plaintext security up to  $\mathcal{O}(2^{n/2})$  queries. They also showed that with an ARCU hash and an Almost Left-Inverse-Collision-avoiding Universal (ALICU) hash, the same 2-round LR construction can achieve security using only 2 random functions. Since then, LR ciphers have been deeply studied in the symmetric-key cryptography literature (see [15] for a full survey).

For over three decades, all designs and analyses of LR constructions have assumed that the internal functions are pseudo-random functions. However, in many practical cases [12,10,1,23], the inner round functions are actually block ciphers, leading to the birthday-bound limitation due to the PRP-PRF switching lemma [3]. A natural alternative is to analyze the LR construction directly using permutations for the round functions. Piret [22] studied the security of permutation-based LR ciphers, but only achieved security up to the birthday bound. Guo et al. [11] managed to go beyond this bound, showing that 3-round, 5-round, and 7-round LR constructions using random permutations achieve known-plaintext (KPA), chosen-plaintext (CPA), and chosen-ciphertext (CCA) security, respectively, up to  $\mathcal{O}(2^{2n/3})$  queries. More recently, Chakraborty et al. [4] extended these results and proved that the same 5-round and 7-round LR constructions with independent random permutations can achieve CPA and CCA security up to  $\mathcal{O}(2^n)$  queries. They also showed how to build optimally secure tweakable PRPs using a 4-round LR construction with independent random permutations and a special class of functions called TLR-compatible function families. Moreover, they showed that tweakable strong PRPs can be constructed using two TLR-compatible functions and a 5-round LR construction with independent random permutations.

## 1.2 Building PRFs from PRPs

As mentioned before, a popular way of constructing PRFs are using PRPs as the underlying primitives. For constant input length PRFs, this can be achieved optimally. Two most popular ways of constructing such PRFs are the Sum of Permutations (SoP) [2] construction along with its single key variant TWIN and the Encrypted Davis-Mayer (EDM) [7] construction. Both of these constructions have been proven to achieve optimal security [19,20,8,7,14].

On the other hand it has been hard to construct variable input length PRFs (VIL-PRFs) and very few constructions could be proven to achieve beyond birthday bound security, most notably, SUM-ECBC construction [24], the PMAC+

construction [25] and its single-key variant 1k-PMAC+ [9], 3kf9 [26] and Light-MAC+ [16], all of which follow the double-block hash-then-PRF paradigm (Db-HtS) [9] and known to achieve suboptimal but beyond the birthday bound security. In Asiacrypt 2020, Cogliati et al. [6], explored this well-known Db-HtS paradigm to construct the Double block Hash-then-Benes construction and showed that this construction can achieve optimal security modulo the assumption and now proven Patarin’s mirror theory result[21,5]. More recently, as an extension of their tweakable PRP result, Chakraborty et al. [4] constructed what they called a DbHtF paradigm and showed that using 4 independent PRPs one can generate an optimally secure VIL-PRF construction.

The above discussion leads to the following questions:

1. Whether the Naor-Reingold approach [17], which reduces the number of primitives by using cheaper hash functions, can be applied to reduce the number of primitive calls in [4]?
2. Given both the Double-block Hash-then-Benes construction and the Double-block Hash-then-Feistel construction use 4 primitive calls, using a construction as above, whether an optimally secure VIL-PRF can be constructed using less than 4 primitive calls?
3. Can the independence relations between the round permutations be relaxed?

### 1.3 Our Contribution

In this paper, we answer all the questions above in the affirmative. To be more specific,

- We start by showing that **PolyHash** not only satisfies AXU property but can also be shown to achieve a much stronger property, which we dubbed as Constant-Sum-universal property (CSU). Then we show that this CSU property of **PolyHash** can be used to construct double-block tweakable keyed hash which satisfies what we dubbed as PCSU property. We further show that this is not a special property only attained by **PolyHash** and such a double-block tweakable keyed hash can also be generated using random permutations.
- Next, we move towards the main objective of the paper and start by showing that the Hash-then-three-round Feistel construction as an  $n$ -bit output VIL-PRF, achieves optimal security.
- In the remaining sections, we extend our proof methodology to extend the results by Naor and Reingold. More specifically, we show that using a family of PCSU secure Double blocked (tweakable) keyed hash family, one can achieve optimal (T)PRP and (T)SPRP secure constructions using 4 and 5 round Feistel construction with random permutations.
- In both the TPRP and TSPRP constructions, we also try to reduce the independency relations amongst the round permutations and show that one can use only three independent random permutations and still retain optimal security.

## 2 Preliminaries

**NOTATIONS:** For two natural numbers  $a$  and  $b$  such that  $a \leq b$ , we write  $[a, b]$  to denote the set  $\{a, a+1, \dots, b\}$ . As a shorthand notation, we write  $[q]$  to denote  $[1, q]$ . For a natural number  $n$ ,  $\{0, 1\}^n$  denotes the set of all binary strings of length  $n$  and  $\{0, 1\}^*$  denotes the set of all binary strings of arbitrary length. For  $X, Y \in \{0, 1\}^n$ , we write  $Z = X \oplus Y$  to denote the xor of  $X$  and  $Y$ . For two strings  $X, Y$ , we write  $(X, Y)$  or  $X \parallel Y$  interchangeably to denote the concatenation of  $X$  followed by  $Y$ . Extending it from one dimension to  $k$  dimension, we write  $(X^k, Y^k)$  to denote the tuple  $((X_1, Y_1), \dots, (X_k, Y_k))$ . We write a  $k$ -tuple  $(X_1, \dots, X_k)$  as  $X^k$ . We call a  $k$ -tuple  $X^k$  *alternating* if for all  $j \in [k-1]$ ,  $X_j \neq X_{j+1}$ . Let  $X^k$  and  $Y^k$  be two  $k$ -tuples, where each  $X_i, Y_i \in \{0, 1\}^n$ , we define  $X^k \oplus Y^k = (X_1 \oplus Y_1, \dots, X_k \oplus Y_k)$  as the co-ordinate wise xor of the two tuples. Let  $X \in \{0, 1\}^{nk}$ . We write  $(X_1, X_2, \dots, X_k) \leftarrow_n X$  to denote that  $X_1, \dots, X_k \in \{0, 1\}^n$  and  $X_1 \parallel \dots \parallel X_k = X$ .

Let  $\mathcal{S}$  be a finite set. For a natural number  $k$ , we write  $\mathcal{S}^k$  to denote the set of  $k$ -tuple  $(S_1, S_2, \dots, S_k)$  where each  $S_i \in \mathcal{S}$ , i.e.,

$$\mathcal{S}^k := \{S^k := (S_1, \dots, S_k) : \forall i \in [k], S_i \in \mathcal{S}\}.$$

By an abuse of notation, we also write  $\mathcal{S}^k$  to denote the set of multiset  $\{S_i : s_i \in \mathcal{S}\}$ . Similarly, we write  $(\mathcal{S})_k$  to denote the set of  $k$ -tuple  $(S_1, S_2, \dots, S_k)$  where each  $S_i \in \mathcal{S}$  and each  $S_i$  are distinct, i.e.,

$$(\mathcal{S})_k := \{S^k := (S_1, \dots, S_k) : \forall i \neq j \in [k], S_i, S_j \in \mathcal{S}, S_i \neq S_j\}.$$

For a given  $k$  tuple  $(X_1, \dots, X_k)$ , where each  $X_i \in \{0, 1\}^n$ , we write  $\max_a |\{i \in [k] : X_i = a\}|$ , where  $a \in \{0, 1\}^n$  to denote the maximum multiplicity of  $X_i$  values. For a given  $n$ -bit string  $X$ , we write  $\lfloor X \rfloor$  to denote the leftmost  $n-1$  bit string of  $X$ .

We write  $X \leftarrow Y$  to denote the assignment of the variable  $y$  into  $x$ . For a random variable  $\mathbf{X}$ ,  $\mathbf{X} \leftarrow_{\$} \{0, 1\}^n$  denotes that  $\mathbf{X}$  is sampled uniformly at random from  $\{0, 1\}^n$ . For a tuple of random variables  $(X_1, \dots, X_q)$ , we write  $(X_1, \dots, X_q) \leftarrow_{\$} \{0, 1\}^n$  to denote that each  $X_i$  is sampled uniformly from  $\{0, 1\}^n$  and independent to all other previously sampled random variables. Similarly, we write  $(X_1, \dots, X_q) \xleftarrow{\text{wor}} \{0, 1\}^n$  to denote that each  $X_i$  is sampled uniformly from  $\{0, 1\}^n \setminus \{X_1, \dots, X_{i-1}\}$ .

Let  $f : \mathcal{X} \rightarrow \{0, 1\}^{2n}$  be a *double block function*. Any such double block function can be realized as a concatenation of two functions  $f^1$  and  $f^2$  where  $f^b : \mathcal{X} \rightarrow \{0, 1\}^n$ , for  $b \in \{1, 2\}$  such that  $f(X) = (f^1(X), f^2(X))$  for each  $X \in \mathcal{X}$ . The set of all functions from  $\mathcal{D}$  to  $\mathcal{R}$  is denoted as  $\text{Func}(\mathcal{D}, \mathcal{R})$ . The set of all permutations over  $\{0, 1\}^n$  is denoted as  $\text{Perm}(n)$ . A random variable  $\text{RF}$  is said to be a random function if  $\text{RF}$  is uniformly distributed over  $\text{Func}(\mathcal{D}, \mathcal{R})$ . Similarly, a random variable  $\text{RP}$  is said to be a random permutation if  $\text{RP}$  is uniformly distributed over  $\text{Perm}(n)$ . For an  $n$ -bit permutation  $\text{P}$  and a bit  $b \in \{0, 1\}$ , we define a function  $\text{P}^b : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^n$  as follows:

$$\text{P}^b(X') = \text{P}(X' \parallel b), \text{ where } X' \in \{0, 1\}^{n-1}.$$

For integers  $1 \leq b \leq a$ , we write  $(a)_b$  to denote  $a(a-1)\dots(a-b+1)$ , where  $(a)_0 = 1$  by convention. A distinguisher  $A$  is an algorithm that tries to distinguish between two oracles  $\mathcal{O}_1$  and  $\mathcal{O}_0$  via black box interaction with one of them. At the end of the interaction, it returns a bit  $b \in \{0, 1\}$ .

## 2.1 Pseudorandom Permutation

Let  $n, \kappa \in \mathbb{N}$  be two natural numbers. A pseudorandom permutation (a.k.a block cipher)  $E : \{0, 1\}^\kappa \times \{0, 1\}^n \rightarrow \{0, 1\}^n$  is a function that takes as input a key  $K \in \{0, 1\}^\kappa$  and an  $n$ -bit string  $x \in \{0, 1\}^n$  and outputs an element  $y \in \{0, 1\}^n$  such that for each  $k \in \{0, 1\}^\kappa$ , the function  $E_k$  is bijective from  $\{0, 1\}^n$  to  $\{0, 1\}^n$ . Due to the bijectivity of the function  $E_k$ , its inverse function  $E_k^{-1}$  exists. We fix positive even integers  $n$  and  $\kappa$  to denote the *block size* and the *key size* of the block cipher, respectively, in terms of number of bits. To capture the security notion of a block cipher  $E$ , we define its pseudorandom permutation (prp) advantage for a distinguisher  $A$  as follows:

$$\text{Adv}_E^{\text{prp}}(A) := |\Pr[A^{E_K} = 1] - \Pr[A^{\text{RP}} = 1]|,$$

where the first probability is calculated over the randomness of  $K \leftarrow \$ \{0, 1\}^\kappa$  and the second probability is calculated over the randomness of  $\text{RP} \leftarrow \$ \text{Perm}(n)$ . We say that  $E$  is  $(q, t, \epsilon)$  secure if the maximum pseudorandom permutation advantage of  $E$  is  $\epsilon$  where the maximum is taken over all distinguishers  $A$  that makes  $q$  queries to its oracle and runs for time at most  $t$ . We say that a block cipher is a strong pseudorandom permutation (sprp) if the adversary  $A$  is given additional access to the inverse of the oracle in both worlds.

## 2.2 (Strong) Tweakable Pseudorandom Permutation

Let  $n, \kappa \in \mathbb{N}$  be two natural numbers. A tweakable pseudorandom permutation (a.k.a tweakable block cipher (TBC)) is a mapping  $\tilde{E} : \{0, 1\}^\kappa \times \mathcal{T} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ , where  $\{0, 1\}^\kappa$  is called the key space and  $\mathcal{T}$  is called the tweak space, such that for all key  $k \in \{0, 1\}^\kappa$  and for all tweak  $\text{Tw} \in \mathcal{T}$ ,  $\tilde{E}_k^{\text{Tw}}$  is a permutation over  $\{0, 1\}^n$ . We denote  $\text{TBC}(\{0, 1\}^\kappa, \mathcal{T}, \{0, 1\}^n)$ , the set of all tweakable block ciphers with key space  $\{0, 1\}^\kappa$ , tweak space  $\mathcal{T}$  and message space  $\{0, 1\}^n$ .

A *tweakable permutation* with tweak space  $\mathcal{T}$  and domain  $\{0, 1\}^n$  is a mapping

$$\tilde{\text{RP}} : \mathcal{T} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$$

such that for all tweak  $\text{Tw} \in \mathcal{T}$ ,  $\tilde{\text{RP}}^{\text{Tw}}$  is a permutation over  $\{0, 1\}^n$ . We write  $\text{TP}(\mathcal{T}, n)$  to denote the set of all tweakable permutations with tweak space  $\mathcal{T}$  and  $n$ -bit messages. We define the advantage of  $A$  in breaking the tweakable pseudorandom permutation (tprp) security of  $\tilde{E}$  as

$$\text{Adv}_{\tilde{E}}^{\text{tprp}}(A) := |\Pr[A^{\tilde{E}_K} = 1] - \Pr[A^{\tilde{\text{RP}}} = 1]|, \quad (1)$$

where the first probability is calculated over the randomness of  $K \leftarrow_{\$} \{0,1\}^\kappa$  and the second probability is calculated over the randomness of  $\widetilde{\text{RP}} \leftarrow_{\$} \text{TP}(\mathcal{T}, n)$ . When the adversary is given access to the tweakable permutation and its inverse in both worlds, then we say the strong tweakable pseudorandom permutation (stprp) advantage of  $\mathbf{A}$  against  $\widetilde{\mathbf{E}}$ . We say that  $\widetilde{\mathbf{E}}$  is  $(q, \mathbf{t}, \epsilon)$ -(s)tprp secure if the maximum (strong) tweakable pseudorandom permutation advantage of  $\widetilde{\mathbf{E}}$  is  $\epsilon$  where the maximum is taken over all distinguishers  $\mathbf{A}$  that makes a total of  $q$  queries to its oracle and runs for time at most  $\mathbf{t}$ .

### 2.3 H-Coefficients Technique

H-Coefficient Technique, introduced by Patarin [?], provides a “systematic” way to upper bound the statistical distance between the answers of two interactive systems and is typically used to prove the information-theoretic pseudo randomness of constructions. Let  $\mathbf{A}$  be a computationally unbounded deterministic distinguisher that interacts with either the real oracle, i.e., the construction of our interest, or the ideal oracle, which is usually considered to be a uniform random function or permutation. The collection of all the queries and responses that  $\mathbf{A}$  made and received to and from the oracle is called the *transcript* of  $\mathbf{A}$ , denoted as  $\tau$ . Sometimes, we allow the oracle to release more internal information to  $\mathbf{A}$  only after  $\mathbf{A}$  completes all its queries and responses, but before it outputs its decision bit. In this case, the transcript of  $\mathbf{A}$  includes the additional information about the oracle, and clearly the maximum distinguishing advantage of  $\mathbf{A}$  in this setting can not be less than that of without additional information. Observe that the transcript  $\tau$  is a random variable and the randomness of the distribution of  $\tau$  only comes from the randomness of the oracle with which  $\mathbf{A}$  interacts.

Let  $\mathbf{X}_{\text{re}}$  denote the random variable that takes a transcript  $\tau$  realized in the real world. Similarly,  $\mathbf{X}_{\text{id}}$  denotes the random variable that takes a transcript  $\tau$  realized in the ideal world. The probability of realizing a transcript  $\tau$  in the ideal (resp. real) world is called *ideal (resp. real) interpolation probability*. A transcript  $\tau$  is said to be attainable with respect to a distinguisher  $\mathbf{A}$ , if its ideal interpolation probability is non-zero. Let  $\Theta$  denote the set of all attainable transcripts. Following these notations, we state the main theorem of the H-Coefficient technique as follows:

**Theorem 1 (H-Coefficient).** *Let  $\mathbf{A}$  be a fixed deterministic distinguisher that has access to either the real oracle  $\mathcal{O}_{\text{re}}$  or the ideal oracle  $\mathcal{O}_{\text{id}}$ . Let  $\Theta = \Theta_{\text{good}} \sqcup \Theta_{\text{bad}}$  be some partition of the set of attainable transcripts  $\Theta$ . For any good attainable transcript  $\tau \in \Theta_{\text{good}}$ , let*

$$\frac{\Pr[\mathbf{X}_{\text{re}} = \tau]}{\Pr[\mathbf{X}_{\text{id}} = \tau]} \geq 1 - \epsilon_{\text{good}},$$

*for some  $\epsilon_{\text{good}} \geq 0$ , and there exists  $\epsilon_{\text{bad}} \geq 0$  such that  $\Pr[\mathbf{X}_{\text{id}} \in \Theta_{\text{bad}}] \leq \epsilon_{\text{bad}}$  holds. Then,*

$$\text{Adv}_{\mathcal{O}_{\text{re}}}^{\mathcal{O}_{\text{id}}}(\mathbf{A}) := |\Pr[\mathbf{A}^{\mathcal{O}_{\text{re}}} = 1] - \Pr[\mathbf{A}^{\mathcal{O}_{\text{id}}} = 1]| \leq \epsilon_{\text{good}} + \epsilon_{\text{bad}}.$$

All the security proofs in the paper have extensively used the framework of H-Coefficients technique, where we have assumed that the adversary  $\mathbf{A}$  is information theoretic, meaning that  $\mathbf{A}$  is computationally bounded but limited with respect to the number of queries it can make. Thus, all the constructions that we have proposed in this paper are based on idealized primitive, e.g., random permutation instead of using a block cipher.

## 2.4 System of Equations Based on Random Permutation Outputs

Consider a system of equations  $\mathbf{A} \cdot \mathbf{Y}^t = \mathbf{\Lambda}^t$  over the Galois field  $\mathbb{F}_{2^n} := \{0, 1\}^n$  where  $\mathbf{A} = (\mathbf{A}_{ij})_{i \in [e], j \in [m]}$  is an  $e \times m$  matrix over  $\mathbb{F}_{2^n}$ ,  $\mathbf{Y} = (Y_1, \dots, Y_m)$  is a  $m$ -tuple of variables and  $\mathbf{\Lambda} := (\lambda_1, \dots, \lambda_e) \in \mathbb{F}_{2^n}^e$ .

**Definition 1.** A solution  $(y_1, \dots, y_m) \in (\{0, 1\}^n)^m$  of the above system of equations  $\mathbf{A} \cdot \mathbf{Y}^t = \mathbf{\Lambda}^t$  is called a pairwise distinct solution (or in short a p.d. solution) if  $z_i$ 's are distinct.

Let  $r$  denote the rank of  $\mathbf{A}$  (in other words, the number of linearly independent columns in  $\mathbf{A}$ ). Then,

$$|\{y^m := (y_1, \dots, y_m) : \mathbf{A} \cdot (y^m)^t = \mathbf{\Lambda}^t, y_i \neq y_j, \forall i \neq j\}| = (2^n)_{m-r}$$

as one can assign an appropriate set of  $m - r$  variables freely with distinct values to the non-basis variables, and the rest  $r$  many basis variables will be determined uniquely (some of the choices may violate the p.d. property, and hence we have an upper bound instead of exact equality). Thus, we have the following lemma

**Lemma 1 (Upper Bound on P.D).** Let  $P$  be an  $n$ -bit random permutation and  $u_1, \dots, u_m \in \{0, 1\}^n$  be distinct elements. Let  $Y_i = P(u_i)$ , for  $i \in [m]$ , and the vector  $\mathbf{Y} = (Y_1, \dots, Y_m)$ . Let  $\mathbf{A}_{m \times e}$  be a rank  $r$  matrix and  $\mathbf{\Lambda} \in \mathbb{F}_{2^n}^e$ . Then, by assuming  $m \leq 2^{n-1}$ , we have

$$\Pr[\mathbf{A} \cdot \mathbf{Y}^t = \mathbf{\Lambda}^t] \leq \frac{(2^n)_{m-r}}{(2^n)_m} \leq \frac{1}{2^{r(n-1)}}. \quad (2)$$

**GRAPH THEORETIC REPRESENTATION OF THE SYSTEM.** Let  $\mathbf{A}$  be a binary (0 or 1 entries) matrix of dimension  $e \times m$  such that each row contains exactly two 1's. Note that each equation in  $\mathbf{A} \cdot \mathbf{Y}^t = \mathbf{\Lambda}^t$  is of the form  $Y_i \oplus Y_j = \lambda_k$  for some  $i, j, k$  with  $i \neq j$ . We call any such system of equations *binary bivariate*. For any binary bivariate system of equations, we can associate a labeled bipartite graph  $G = (V := [m], E, L)$  where the edge set  $E = \{\{j, k\} \mid \exists i \in [m] \ni A_{ij} = A_{ik} = 1\}$  and  $L(j, k) = \lambda_i$ . Thus,  $|E| = e$ . In other words, every labeled edge corresponds to an equation, and every node corresponds to a variable. For any path  $P$  (or cycle  $C$ ) we define  $\lambda(P)$  (resp.  $\lambda(C)$ ) to be the xor of the labels of the edges present in the path (resp. cycle). It is easy to see that the rank of  $A$  is the same as  $m - c$  where  $c$  is the number of components, where each component with  $m'$  nodes contributes exactly  $m' - 1$  ranks. It is crucial to see that if there is a cycle  $C$  with  $\lambda(C) \neq 0^n$  or there is a path  $P$  with  $\lambda(P) = 0^n$  then the system

of equations  $\mathbf{A} \cdot \mathbf{Y}^t = \mathbf{A}^t$  cannot have any pairwise distinct solution. We call a system of binary bivariate equations *nice*, if the associated labeled graph  $G$  satisfies the following properties:

1.  $G$  does not contain any cycle
2.  $G$  does not contain any even-length path such that the label sum of the path is zero.

Having defined the nice system of equations, we are interested to obtain a lower bound on the number of p.d solutions to a given system of nice equations. This result is popularly known as *mirror theory*.

**Lemma 2 (Lower Bound on P.D).** *Let  $\mathbb{E}$  be a nice system of equations over  $\{0, 1\}^n$  and  $G_{\mathbb{E}} = (V_1 \sqcup V_2, E)$  be the associated edge-labeled bipartite graph for the system of equations  $\mathbb{E}$ . Let the number of edges in  $G_{\mathbb{E}}$  is  $q$  and the size of the largest component in  $G_{\mathbb{E}}$  is  $\xi_{\max}$ . If  $\xi_{\max}^2 n + \xi_{\max} \leq 2^{n/2}$  and  $q\xi_{\max}^2 \leq 2^n/12$ , then the number of solutions to  $\mathbb{E}$ , denoted as  $h(\mathbb{E})$  is*

$$h(\mathbb{E}) \geq \frac{(2^n - 2)^{|V_1|}(2^n - 2)^{|V_2|}}{2^{nq}}.$$

### 3 Security Definition of Keyed Hash Function Revisited

In this section, we revisit some useful and existing security definitions of keyed hash function and introduce an another security notion, called *constant sum universal* property. We first show that the constant sum universal property of a keyed hash function is a stronger security notion than the folklore universal property. We also showcase our result by showing that the polyhash function achieves the constant sum universal property. Followed by, we introduce another primitive called *tweakable keyed hash function*, which is a hash function that takes a tweak along with a message  $M \in \mathcal{M}$  such that given the tweak, the hash function is invertible over the message space  $\mathcal{M}$ . We give two concrete instantiations of tweakable keyed hash function and showed that both of them achieves *partial constant sum universal* property.

**Definition 2 (Universal and Almost-XOR-Universal Property).** *A keyed hash function  $H: \mathcal{K}_{\text{hash}} \times \mathcal{M} \rightarrow \mathcal{R}$  is called an  $\epsilon$ -universal hash function, if  $\forall M_1 \neq M_2 \in \mathcal{M}$ , we have*

$$\Pr[K \leftarrow \mathcal{K}_{\text{hash}} : H_K(M_1) = H_K(M_2)] \leq \epsilon.$$

*A keyed hash function  $H: \mathcal{K}_{\text{hash}} \times \mathcal{M} \rightarrow \{0, 1\}^n$  is called an  $\epsilon$ -AXU hash function, if  $\forall M_1 \neq M_2 \in \mathcal{M}$  and  $\forall \Delta \in \{0, 1\}^n$ , we have*

$$\Pr[K \leftarrow \mathcal{K}_{\text{hash}} : H_K(M_1) \oplus H_K(M_2) = \Delta] \leq \epsilon.$$



**Definition 3 (Zero-Sum Universal Property).** A keyed hash function  $H: \mathcal{K}_{\text{hash}} \times \mathcal{M} \rightarrow \{0,1\}^n$  is called an  $\delta$ -Zero-Sum Universal (ZSU) hash function, if there exists a function  $f: \bigcup_{k=2}^{\infty} \mathcal{M}^k \rightarrow \mathcal{M}$  such that for all  $\ell \geq 2$  and distinct  $M_1, \dots, M_{\ell-1} \in \mathcal{M}$  with  $M_{\ell} \neq f(M_1, \dots, M_{\ell-1})$ ,

$$\Pr[K \leftarrow \mathcal{K}_{\text{hash}} : H_K(M_1) \oplus \dots \oplus H_K(M_{\ell}) = 0^n] \leq \delta.$$

**Definition 4 (Constant-Sum Universal Property).** A keyed hash function  $H: \mathcal{K}_{\text{hash}} \times \mathcal{M} \rightarrow \{0,1\}^n$  is called an  $\delta$ -Constant-Sum Universal (CSU) hash function, if for any  $\Delta \in \{0,1\}^n$  there exists a function  $f^{\Delta}: \bigcup_{k=2}^{\infty} \mathcal{M}^k \rightarrow \mathcal{M}$  such that for all  $\ell \geq 2$  and distinct  $M_1, \dots, M_{\ell-1} \in \mathcal{M}$  with  $M_{\ell} \neq f^{\Delta}(M_1, \dots, M_{\ell-1})$ ,

$$\Pr[K \leftarrow \mathcal{K}_{\text{hash}} : H_K(M_1) \oplus \dots \oplus H_K(M_{\ell}) = \Delta] \leq \delta.$$

By definition, the constant-sum universal property implies the zero-sum universal property. However, we do not know whether the reverse is true or not. In the following, we give a few concrete examples of the constant-sum universal hash function.

### 3.1 Polyhash is a Constant-Sum Universal Hash Function

We first define for any  $c \in \{0,1\}^n$ , the  $\text{add}_c$  function (precisely, this function adds  $c$  to the last  $n$ -bit of the input) from  $\{0,1\}^n$  to  $\{0,1\}^n$  such that

$$\text{add}_c(X) = X \oplus 0^{|X|-n} \| c.$$

We next define the  $\text{pad}$  function. Let  $X \in \{0,1\}^*$ . We define a function  $\text{pad} : \{0,1\}^* \rightarrow (\{0,1\}^n)^+$  to be a bijective padding function defined as follows.

$$\text{pad}(X) = \begin{cases} X \| 10^{n-1-(|X| \bmod n)} & \text{if } |X| \bmod n \neq 0 \\ X \| 10^{n-1} & \text{else} \end{cases}$$

We write  $\text{pad}^{-1}$  to denote the inverse function. Given  $X_0, \dots, X_{t-1} \in \{0,1\}^n$ , we define the following polynomial the indeterminate  $K$  as follows:

$$\text{Poly}_K(X_0, \dots, X_{t-1}) = X_{t-1} \cdot K^{t-1} \oplus X_{t-2} \cdot K^{t-2} \oplus \dots \oplus X_1 \cdot K \oplus X_0.$$

For  $X \in \{0,1\}^*$ , we define the classical poly-hash function:

$$\text{PolyHash}_K(X) = \text{Poly}_K(X_0, \dots, X_{t-1})$$

where  $(X_1, \dots, X_{t-1}) \leftarrow_n \text{pad}(X)$ . By the definition of the padding function  $\text{pad}$ , the last  $n$ -bit is non-zero and hence the degree of the polynomial is exactly  $t-1$ . Now, we state some simple observations on the poly-hash function.

1. **Concatenation Property:** For all  $L \in \{0,1\}^n$ ,  $\text{PolyHash}_K(L \| X) = L \oplus K \cdot \text{PolyHash}_K(X)$ .

2. **Additive Property:** For all  $X_1, \dots, X_t \in (\{0, 1\}^n)^+$ , we have

$$\bigoplus_{i=1}^t \text{Poly}_K(X_i) = \text{Poly}_K(X_1 \oplus \dots \oplus X_t).$$

Let  $X_i = \text{pad}(M_i)$  for  $M_1, \dots, M_{t-1}$ . For any  $\Delta$ , let  $M_t^\Delta$  be the unique bit string such that  $\text{pad}(M_t^\Delta) = \text{add}_\Delta(X_1 \oplus X_2 \oplus \dots \oplus X_{t-1})$ , denoted as  $M_t^\Delta = f_{\text{poly}}^\Delta(M_1, \dots, M_{t-1})$ . Then, we have

$$\text{PolyHash}_K(M_t^\Delta) = \bigoplus_{i=1}^{t-1} \text{PolyHash}_K(M_i) \oplus \Delta.$$

**Lemma 3 (CSU Property of PolyHash).** *PolyHash<sub>K</sub> defined over bit strings of size less than  $n\ell$  is  $(\ell/2^n)$ -CSU.*

*Proof.* We consider for all  $\Delta$ ,  $f_{\text{poly}}^\Delta$  as defined above for proving the CSU property. Fix a  $\Delta \in \{0, 1\}^n$ . Let  $M_1, M_2, \dots, M_t$  be distinct inputs such that,  $M_t \neq f_{\text{poly}}^\Delta(M_1, \dots, M_{t-1})$ . Hence,  $\text{pad}(M_t) \neq \text{add}_\Delta(X_1 \oplus X_2 \oplus \dots \oplus X_{t-1})$ . Thus, we have

$$\bigoplus_{i=1}^t \text{PolyHash}_K(M_i) = \bigoplus_{i=1}^t \text{Poly}_K(X_i) = \text{Poly}_K(X_1 \oplus \dots \oplus X_t)$$

By choice of  $M_t$ , we know that  $X_1 \oplus \dots \oplus X_t \neq * \parallel \Delta$  and the result follows.  $\square$

### 3.2 Tweakable Keyed Hash Function and Partial Constant (Zero)-Sum-Universal

In this section we introduce a variant of a keyed hash function, called *tweakable keyed hash function*. It is similar to the usual keyed hash function, except that it takes an input tweak in addition to the regular message. Moreover, given the tweak, the hash function is invertible. Let  $H : \mathcal{K}_{\text{hash}} \times \mathcal{T} \times \mathcal{D} \rightarrow \mathcal{D}$  be a keyed tweakable hash function, where  $\mathcal{T}$  is the tweak space, such that for all  $K \in \mathcal{K}_{\text{hash}}$  and for all tweak  $\text{Tw} \in \mathcal{T}$ ,  $H_K(\text{Tw}, \cdot)$  is a permutation over  $\mathcal{D}$ . In this paper, we consider  $\mathcal{D} = \{0, 1\}^{2n}$  and in that case, we call the tweakable keyed hash function a *double-block tweakable keyed hash function*.

**Definition 5.** Let  $H = (H^1, H^2) : \mathcal{K}_{\text{hash}} \times \{0, 1\}^* \rightarrow \{0, 1\}^{2n}$  be a double-block keyed hash function. We call  $H$  to be an  $(\epsilon, \delta)$ -partial constant (zero)-sum-universal, if

1.  $H$  is  $\epsilon$ -universal, and
2.  $H^1$  is  $\delta$ -constant (zero)-sum-universal.

In the following, we provide two simple instantiations of a double-block tweakable keyed hash function that achieves a partial constant-sum-universal property.

### 3.3 Instantiations of Double-Block Tweakable Keyed Hash

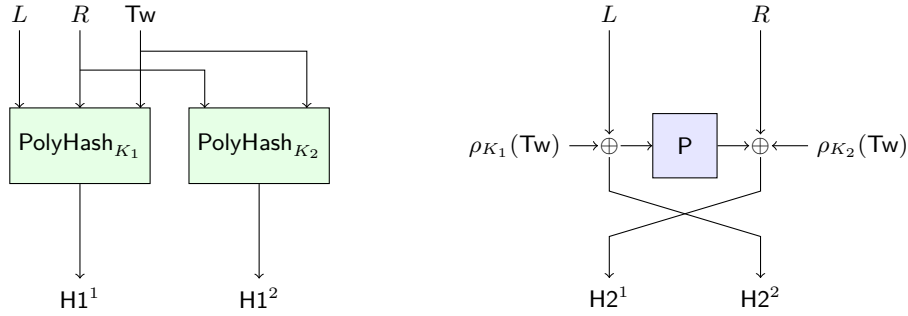
In this section, we propose two concrete instantiations of a double block tweakable keyed hash function, one is based on the polyhash function and the other one is based on random permutation. We show that both constructions achieve the partial constant-sum-universal property.

**POLY-HASH BASED DOUBLE BLOCK TWEAKABLE KEYED HASH.** For hash key  $K_1, K_2 \in \{0, 1\}^n$ , and an input  $L \| R \| \text{Tw}$  where  $L, R \in \{0, 1\}^n$ ,  $\text{Tw} \in \{0, 1\}^*$ , we define a  $2n$ -bit tweakable keyed hash function  $H$  as follows:

$$H1_{K_1, K_2}(L, R, \text{Tw}) = (H1^1 = \text{Poly-Hash}_{K_1}(L \| R \| \text{Tw}), H1^2 = \text{Poly-Hash}_{K_2}(R \| \text{Tw})).$$

A pictorial description of the hash function has been shown in the left part of Fig 3.3. Note that, for a given tweak  $\text{Tw}$ , the function  $H1$  is invertible. The following result shows that  $H1$  is a partial constant-sum-universal tweakable keyed hash function, proof of which is deferred to Section 3.4.

**Lemma 4.** *The hash function  $H1$  as defined above over bit strings of size less than  $\ell n$  is  $(\ell^2/2^{2n}, \ell/2^n)$ -partial-CSU tweakable keyed hash function.*



**Figure 1.** PZSU Hash Functions:  $H1$  and  $H2$ . The function  $\rho : \{0, 1\}^* \rightarrow \{0, 1\}^n$  is an AXU-hash and  $P$  is a random permutation.

**RANDOM PERMUTATION BASED DOUBLE BLOCK TWEAKABLE KEYED HASH.** Let  $P$  be an  $n$ -bit random permutation and  $\rho$  be an  $\epsilon$ -almost-xor-universal hash function, defined over the tweak space  $\mathcal{T}$ . For hash key  $K_1, K_2 \in \{0, 1\}^n$ , and an input  $L \| R \| \text{Tw}$  where  $L, R \in \{0, 1\}^n$ ,  $\text{Tw} \in \{0, 1\}^*$ , we now define a  $2n$ -bit hash function  $H2$  as follows:

$$H2_{K_1, K_2, P}(L, R, \text{Tw}) = (H2^1 = L \oplus \rho_{K_2}(\text{Tw}) \oplus P(H2^2), H2^2 = \rho_{K_1}(\text{Tw}) \oplus R).$$

A pictorial description of the hash function has been shown in the right part of Fig 3.3. Note that, for a given tweak  $\text{Tw}$ , the function  $H2$  is invertible. The following result shows that  $H2$  is a partial constant-sum-universal tweakable keyed hash function, proof of which is deferred to Section 3.5.

**Lemma 5.** *The hash function H2 as defined above over bit strings of size less than  $\ell n$  is  $(\epsilon^2, \epsilon + 2^{-n+1})$ -partial-CSU tweakable keyed hash function.*

### 3.4 Proof of Lemma 4

First, we show H1 is  $\ell^2/2^{2n}$  universal. Let  $L\|R\|\text{Tw} \neq L'\|R'\|\text{Tw}'$ . Now if  $R\|\text{Tw} = R'\|\text{Tw}'$  then

$$\begin{aligned} \text{Poly-Hash}_{K_1}(L\|R\|\text{Tw}) &= L \oplus K_1 \cdot \text{Poly-Hash}_{K_1}(R\|\text{Tw}) \\ &\neq L' \oplus K_1 \cdot \text{Poly-Hash}_{K_1}(R'\|\text{Tw}') \\ &= \text{Poly-Hash}_{K_1}(L'\|R'\|\text{Tw}'). \end{aligned}$$

Thus, we assume that  $R\|\text{Tw} \neq R'\|\text{Tw}'$ . Now, by using the independence of H1<sup>1</sup> and H1<sup>2</sup>, the result follows. Now we prove that H1 defined over bit strings of size less than  $\ell n$  is  $\ell/2^n$ -CSU. This follows directly from Lemma 3. Finally, we show that the function is invertible. Note that  $R$  is the constant term of  $\text{Poly}(K_2, R\|\text{Tw})$ . Hence, given  $\text{Poly}(K_2, R\|\text{Tw})$ ,  $\text{Tw}$ , and  $K_2$ ,  $R$  can be uniquely retrieved. Similarly, using this uniquely retrieved  $R$ , and given  $\text{Poly}(K_1, L\|R\|\text{Tw})$ ,  $\text{Tw}$ , and  $K_1$ , we can uniquely retrieve  $L$  ( $L$  is the constant term of  $\text{Poly}(K_1, L\|R\|\text{Tw})$ ).  $\square$

### 3.5 Proof of Lemma 5

We first show H2 is  $\epsilon^2$  universal. Let  $L\|R\|\text{Tw} \neq L'\|R'\|\text{Tw}'$ . Now if  $R\|\text{Tw} = R'\|\text{Tw}'$  then  $L \neq L'$ , and

$$L \oplus \rho_{K_2}(\text{Tw}) \oplus P(\rho_{K_1}(\text{Tw}) \oplus R) \neq L' \oplus \rho_{K_2}(\text{Tw}') \oplus P(\rho_{K_1}(\text{Tw}') \oplus R'),$$

Thus, we assume that  $R\|\text{Tw} \neq R'\|\text{Tw}'$ . In this case, if  $\text{Tw} = \text{Tw}'$ , then

$$\rho_{K_1}(\text{Tw}) \oplus R \neq \rho_{K_1}(\text{Tw}') \oplus R'.$$

Thus, assuming  $\text{Tw} \neq \text{Tw}'$ , the probability of collision in H2<sup>2</sup> can be bounded by  $\epsilon$ . Note that a collision in H2<sup>2</sup> implies collision in  $P(\text{H2}^2)$ . Hence, the probability of collision in H2<sup>1</sup> given collision in H2<sup>2</sup> is upper bounded by  $\epsilon$  (as the  $P(\text{H2}^2)$  values cancel out).

We now show  $\text{H2}_{K_1, K_2, P}^1(\cdot)$  is  $(\epsilon + 2^{-n+1})$ -CSU hash up to  $2^{n-1}$  summations. Fix a  $\Delta \in \{0, 1\}^n$ . We define  $f_{\text{H2}^1}^\Delta$  as follows for all  $\ell \geq 2$ .

$$f_{\text{H2}^1}^\Delta((L_1, R_1, \text{Tw}_1), \dots, (L_\ell, R_\ell, \text{Tw}_\ell)) = (\oplus_{i=1}^\ell L_i \oplus \Delta, \oplus_{i=1}^\ell R_i, \oplus_{i=1}^\ell \text{Tw}_i).$$

We now prove the CSU bound. Let  $(L_1, R_1, \text{Tw}_1), \dots, (L_t, R_t, \text{Tw}_t)$  be distinct inputs such that

$$(L_t, R_t, \text{Tw}_t) \neq f_{\text{H2}^1}^\Delta((L_1, R_1, \text{Tw}_1), \dots, (L_{t-1}, R_{t-1}, \text{Tw}_{t-1})).$$

We have to show that,

$$\Pr [\oplus_{i=1}^t \text{H2}_{K_1, K_2, P}^1(L_i, R_i, \text{Tw}_i) = \Delta] \leq \epsilon + 2^{-n+1}$$

We will show this bound following the cases below.

- **Case 1:**  $\text{Tw}_t \neq \oplus_{i=1}^{t-1} \text{Tw}_i$  or  $R_t \neq \oplus_{i=1}^{t-1} R_i$ . Hence, there is at least one  $(R, \text{Tw})$  pair that appears an odd times among  $(R_1, \text{Tw}_1), \dots, (R_t, \text{Tw}_t)$ . In that case, we get the term  $P(\rho_{K_1}(\text{Tw}) + R)$ , that does not get canceled out with the other  $P(\rho_{K_1}(\text{Tw}_i) + R_i)$ s. We now show the ZSU bound following the two subcases.
  - In the first subcase, we get exactly one  $(R, \text{Tw})$  pair that appears an odd times. In this case, we get the bound ZSU bound from the entropy of  $P$  (from  $P(\rho_{K_1}(\text{Tw}) + R)$ ), which is  $2^{-n}$ .
  - In the second subcase, we get at least two  $(R, \text{Tw})$  pairs (say  $t \geq d \geq 2$  pairs) that appear odd times. Hence, there can be two cases. In the first one, all the  $d$   $(R, \text{Tw})$  pairs collide to the same  $\rho_{K_1}(\text{Tw}) + R$  value. Since we need at least one collision, this probability can be bounded by  $\epsilon$ . In the second, no collision in  $\rho_{K_1}(\text{Tw}) + R$ . In the worst case, for all the  $d$   $(R, \text{Tw})$  pairs,  $\rho_{K_1}(\text{Tw}) + R$  are distinct. Since,  $t \geq d$ , this probability is upper bounded by  $\frac{1}{2^{n-t+1}}$ . Thus, the CSU-bound is  $\epsilon + \frac{1}{2^{n-t+1}}$ .
- **Case 2:**  $\text{Tw}_t = \oplus_{i=1}^{t-1} \text{Tw}_i$  and  $R_t = \oplus_{i=1}^{t-1} R_i$ . We compute the bound following the two subcases.
  - In the first subcase, all the  $(R, \text{Tw})$  pairs among  $(R_1, \text{Tw}_1), \dots, (R_t, \text{Tw}_t)$  occur even times. In this case,

$$\oplus_{i=1}^t \rho_{K_2}(\text{Tw}_i) \oplus \oplus_{i=1}^t P(\rho_{K_2}(\text{Tw}_i) \oplus R_i) = 0^n$$

As  $\oplus_{i=1}^t L_i \neq \Delta$ , the CSU bound is 0.

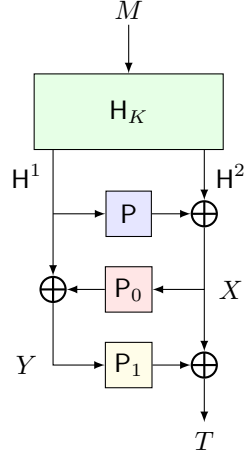
- In the second subcase, there is at least one  $(R, \text{Tw})$  pair that appears an odd times among  $(R_1, \text{Tw}_1), \dots, (R_t, \text{Tw}_t)$  (take the example of  $t = 4$ , and the  $(R, \text{Tw})$  pairs are  $(R_1, \text{Tw}_1)$ ,  $(R_2, \text{Tw}_2)$ ,  $(R_1, \text{Tw}_2)$ , and  $(R_2, \text{Tw}_1)$ ). In this case, following exactly the same way as in the second subcase of Case 1, we get the ZSU bound as  $\epsilon + \frac{1}{2^{n-t+1}}$ .

Thus, for  $t \leq 2^{n-1}$ , we get the required CSU bound up to  $2^{n-1}$  summations.

The invertibility proof is straightforward, as H2 has a similar structure to the single-round Luby-Rackoff construction.  $\square$

## 4 Hash-then-Three round Feistel: An Optimally Secure VIL PRF

In this section, we propose an optimally secure variable input length PRF, called HF<sup>3</sup>, which is a composition of an almost universal  $2n$ -bit hash function with a permutation-based three-round Feistel construction, chopping out the left half value of the construction. It takes an arbitrary length message  $M \in \{0, 1\}^*$  which is fed through a  $2n$ -bit keyed hash function. Let the output be  $(H^1, H^2)$ , which becomes the input of a three-round permutation-based Feistel construction, where all the permutations have been assumed to be sampled independently from the set of all  $n$ -bit permutations. We chop the left half value of the Feistel and return its right half value as the output. The construction is depicted in Fig. 2 and formally presented in Algorithm 1.



**Figure 2.**  $\text{HF}^3$ : Hash-then-Feistel Construction.  $H$  is an  $2n$ -bit almost universal keyed hash function,  $P$ ,  $P_0$ , and  $P_1$  are three independent  $n$ -bit permutations.

---

**Algorithm 1** Hash-then-Three round Feistel Construction.

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```

1: function  $\text{HF}^3[H, P, P_0, P_1](K, M)$ 
2:    $(H^1, H^2) \leftarrow_n H_K(M)$ 
3:    $X \leftarrow P(H^1) \oplus H^2$ 
4:    $Y \leftarrow P_0(X) \oplus H^1$ 
5:    $T \leftarrow P_1(Y) \oplus X$ 
6:   return  $T$ 

```

---

In the following, we show that  $\text{HF}^3$  achieves optimal PRF security. Formally, we have the following result

**Theorem 2.** *Let  $H$  be a  $(\epsilon, \delta)$ - $2n$ -bit partial zero-sum-universal keyed hash function. Let  $P$ ,  $P_0$ , and  $P_1$  be three independently sampled  $n$ -bit permutations. Then, for any information theoretic deterministic distinguisher  $A$  that makes  $q$  queries to  $\text{HF}^3[H, P, P_0, P_1]$ , one has*

$$\text{Adv}_{\text{HF}^3[H, P, P_0, P_1]}^{\text{prf}}(A) \leq \frac{16nq}{2^n} + \left(\frac{4q}{2^n}\right)^n + \frac{12q}{2^n} + \left(\frac{2q}{2^n}\right)^n + \frac{q}{2^n} + \binom{q}{2} \left(\epsilon + \frac{\delta}{2^n}\right) + q\delta,$$

assuming  $qn^2 \leq 2^n/12$  and  $n \geq 29$ .

#### 4.1 Proof of Theorem 2

Let  $A$  be a computationally unbounded deterministic distinguisher that interacts with the oracle  $\text{HF}^3$  in the real world or with an  $n$ -bit uniform random function RF in the ideal world. This is known as the online phase. After the interaction,

the oracle releases some additional information to the distinguisher: In the ideal world, the oracle samples a dummy hash key  $K$  and an  $n$ -bit random permutation  $P$  independently at random. Then, it computes

$$(H_i^1, H_i^2) \leftarrow H_K(M_i), \quad X_i = P(H_i^1) \oplus H_i^2, \quad i \in [q].$$

Similarly, in the real world, the oracle computes using the hash key and the random permutation, which were sampled in the online phase.

ATTACK AND INNER TRANSCRIPT. In both worlds, the attack transcript (the view of the distinguisher after the offline phase is over) is denoted as

$$\tau = ((M_1, T_1, H_1^1, H_1^2, X_1), \dots, (M_q, T_q, H_q^1, H_q^2, X_q)).$$

For any  $i, j \in [q]$ ,  $b \in \{1, 2\}$ , we define

$$\lambda(i, j, b) = \begin{cases} H_i^2 \oplus H_j^2 & \text{if } b = 1 \\ H_i^2 \oplus H_j^2 \oplus T_i \oplus T_j & \text{if } b = 2 \end{cases}$$

The inner transcript (which is a part of the attack transcript) is defined as

$$\tau' := \tau'(T^q, K) := ((M_1, T_1, H_1^1, H_1^2), \dots, (M_q, T_q, H_q^1, H_q^2)).$$

Note that,  $\tau'$  is defined over only the tuple  $T^q$  and the hash key  $K$ .

ATTACK TRANSCRIPT IN THE REAL WORLD. Now we investigate the relationship among the values present in the attack transcript generated in the real world. We first note that

$$P_0(X_i) \oplus P_1^{-1}(X_i \oplus T_i) = H_i^1.$$

Note that  $X_i = X_j$  is equivalent in saying that

$$P(H_i^1) \oplus P(H_j^1) = \lambda(i, j, 1).$$

Similarly, the equation  $X_i \oplus T_i = X_j \oplus T_j$  implies that

$$P(H_i^1) \oplus P(H_j^1) = \lambda(i, j, 2).$$

SYSTEM OF LINEAR EQUATIONS. Let  $\{P_h : h \in \{0, 1\}^n\}$  be a collection of  $n$ -bit variables. Let  $G = (V \subseteq [q], E, B)$  be a  $\{1, 2\}$ -labeled graph. For any edge  $e$  between  $i$  and  $j$  with a label  $b := B(e)$ , we associate an equation

$$\mathbb{E}_{\tau'}(i, j, b) : P_{H_i^1} \oplus P_{H_j^1} = \lambda(i, j, b).$$

We call this graph a *query graph*. If it is a path or a cycle graph, then we call it *query path graph* or *query cycle graph*, respectively. The collection of all equations over the set of all edges of the query graph  $G$  is denoted as  $\mathbb{E}_{\tau'}(G)$ .

**Definition 6.** Let  $i^{k+1} := (i_1, \dots, i_{k+1})$  be an alternating tuple, where each  $i_j \in [q]$ . Let  $G$  be a labeled walk over the sequence  $i^{k+1}$  with a label sequence  $b^k$ , denoted as  $(i^{k+1}, b^k)$ . We call the labeled walk alternating if  $b^k$  is an alternating sequence and  $b_k \neq b_1$  whenever  $i_{k+1} = i_1$  (i.e., when the walk is a loop).

Thus, the system of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  consists of the equations  $\mathbb{E}_{\tau'}(i_1, i_2, b_1)$ ,  $\mathbb{E}_{\tau'}(i_2, i_3, b_2)$ ,  $\dots$ ,  $\mathbb{E}_{\tau'}(i_k, i_{k+1}, b_k)$ .

**Definition 7.** Let  $G$  be a labeled query graph. We say that  $G$  is realizable by an attack transcript  $\tau$  if for all edge  $\{i, j\}$  in  $G$  with a label  $b$ , we have

$$\begin{cases} X_i = X_j & \text{if } b = 1 \\ X_i \oplus T_i = X_j \oplus T_j & \text{if } b = 2 \end{cases}$$

**Lemma 6.** Let  $G$  be a labeled query graph over a vertex set  $V \subseteq [q]$  with label set  $\{1, 2\}$ . If  $\tau$  realizes a labeled graph  $G = (V, E, B)$  then  $(P_h = P(h), \forall h \in \{0, 1\}^n)$  is a solution for the system  $\mathbb{E}_{\tau'}(G)$ .

*Proof.* For all label 1 edges of  $G$ , it holds that  $X_i = X_j$  is same as

$$P(H_i^1) \oplus P(H_j^1) = \lambda(i, j, 1).$$

Similarly, for all label 2 edges of  $G$ , it holds that  $X_i \oplus T_i = X_j \oplus T_j$  is same as

$$P(H_i^1) \oplus P(H_j^1) = \lambda(i, j, 2).$$

Hence,  $P_{H_i^1} = P(H_i^1)$  is a solution for the system of equations  $\mathbb{E}_{\tau'}(G)$ .  $\square$

**FIRST HASH COLLISION (FHC).** For any  $k \geq 1$ , we define the following set  $\mathcal{I}^{(k)} := \{i^{k+1} \in [q]^{k+1} \mid i_1, \dots, i_{k+1}, H_{i_1}^1, \dots, H_{i_k}^1 \text{ are distinct}, H_{i_1}^1 = H_{i_{k+1}}^1\}$ .

**Definition 8.** We say that the first hash-collision (FHC) event holds if there exists an  $k \geq 1$  such that  $i^{k+1} \in \mathcal{I}^{(k)}$  and the attack transcript  $\tau$  realizes an alternating labeled query path graph ( $i^{k+1}, b^k := (b_1, \dots, b_k)$ ) and the label sum of the path is zero, i.e.,

$$\lambda(i^{k+1}, b^k) := \bigoplus_{j=1}^k \lambda(i_j, i_{j+1}, b_j) = 0^n.$$

**Definition 9.** We call an inner transcript  $\tau'$  to be **bad** if one of the following conditions holds:

- **Multi-Coll** :  $\max_a |\{i : T_i = a\}| \geq n$ ;
- **Many-Coll** :  $|\{(i, j) : T_i = T_j, i \neq j\}| \geq q$ ;
- **Univ** :  $\exists i \neq j \in [q] : H_K(M_i) = H_K(M_j)$ ;
- **Cross-Coll-Univ** :  $\exists i, j \in [q]$  such that  $H_i^1 = H_j^1 \wedge H_i^2 \oplus T_i = H_j^2 \oplus T_j$ .
- **Zero-Hash-Sum** : For some  $k \geq 2$ ,

$$|\{(i_1, i_2, \dots, i_k) \in ([q])_k : H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = 0^n\}| \geq 3q^{k-1};$$



Let us define an event `inner_bad` as the union of the above events, i.e.,

$$\text{inner\_bad} := (\text{Mult\_Coll} \vee \text{Many\_Coll} \vee \text{Cross\_Coll\_Univ} \vee \text{Univ} \vee \text{Zero\_Hash\_Sum}).$$

Note that, if an inner transcript  $\tau'$  is bad then the event `inner_bad` event holds true.

**Lemma 7.**

$$\Pr[\text{inner\_bad}] \leq \left(\frac{2q}{2^n}\right)^n + \frac{q}{2^n} + \binom{q}{2} \delta_{\text{univ}} + \binom{q}{2} \frac{\delta_{\text{csu}}}{2^n} + q\delta_{\text{csu}}.$$

*Proof.* We individually upper bound the probability of the event, and then we combine them by virtue of the union bound to yield the result.

**Bounding Mult-Coll:** For a fixed choices of indices  $i_1, i_2, \dots, i_n$ , the event  $T_{i_1} = T_{i_2} = \dots = T_{i_n}$  holds with probability  $2^{-n(n-1)}$ , as there are  $n-1$  equalities and each such equality contributes  $2^{-n}$  to the probability of the event. Moreover, the number of possible choices of indices is at most  $q^n$ . Thus, we have

$$\Pr[\text{Mult-Coll}] \leq \frac{q^n}{2^{n(n-1)}} = \left(\frac{2q}{2^n}\right)^n. \quad (3)$$

**Bounding Many-Coll:** Expected number of collision pair is at most  $q^2 \cdot 2^{-n}$ . Hence, by using Markov inequality,  $\Pr[\text{Many-Coll}] \leq \frac{q}{2^n}$ .

**Bounding Univ, and Cross-Coll-Univ:** For a fixed choices of indices  $i \neq j$ , the event  $H_K(M_i) = H_K(M_j)$  holds with probability  $\epsilon$ . By varying over all possible choices of indices  $i \neq j$ , we have

$$\Pr[\text{Univ}] \leq \binom{q}{2} \epsilon. \quad (4)$$

To bound the probability of the `Cross-Coll-Univ` event, let us assume  $i < j$ . For a fixed choices of indices  $i < j$ , we bound the probability of the event  $H_i^1 = H_j^1$  and  $H_i^2 \oplus T_i = H_j^2 \oplus T_j$ . By conditioning on  $T_i$  and  $M_j$ , the event  $H_i^1 = H_j^1$  holds with probability  $\delta$  (due to the universal property of the  $H^1$  function). Once we fix the hash key,  $H_i^2 \oplus H_j^2$  gets also fixed. Thus, the event  $T_j = T_i \oplus H_i^2 \oplus H_j^2$  holds with probability  $2^{-n}$  (due to the randomness of  $T_j$ ). Note that we already have conditioned  $T_i$  and hence we cannot use its entropy to bound the event. By varying over all possible choices of indices  $i \neq j$ , we have

$$\Pr[\text{Cross-Coll-Univ}] \leq \binom{q}{2} \delta \cdot 2^{-n}. \quad (5)$$

**Bounding Zero-Hash-Sum:** Let  $X$  denotes the cardinality of the set  $\{(i_1, i_2, \dots, i_k) : H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = 0^n\}$ . For each tuple  $(i_1, \dots, i_k) \in ([q])_k$ , we define an indicator random variable  $X_{i_1, \dots, i_k}$  as follows: it takes the value 1 if  $H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = 0^n$  and 0 otherwise. Then, we have

$$X = \sum_{i_1, \dots, i_k} X_{i_1, \dots, i_k}.$$

Therefore, by linearity of expectations, we have

$$\mathbf{Exp}[X] = (q)_k \Pr[\mathbf{H}_{i_1}^1 \oplus \dots \oplus \mathbf{H}_{i_k}^1 = 0^n] \leq q^k \delta.$$

Therefore, by applying Markov's inequality, we have

$$\Pr[\text{Zero\_Hash\_Sum}] = \Pr[X > 3q^{k-1}] \leq \frac{\mathbf{Exp}[X]}{3q^{k-1}} \leq q\delta.$$

Note that the event `Zero_Hash_Sum` for  $k = 2$  is the same as saying that the number of colliding pairs of  $\mathbf{H}^1$  values is at least  $3q$ .  $\square$

**Lemma 8.**  $\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \frac{12nq}{2^n}.$

*Proof.* For any  $k \geq 1$ , we define the event  $\text{FHC}_k$  as follows: we say that  $\text{FHC}_k$  holds if (i)  $i^{k+1} \in \mathcal{I}^{(k)}$ , and (ii)  $\tau$  realizes an alternating label path  $(i^{k+1}, b^k)$ , and (iii) the label sum of the path is zero. Thus,

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \Pr[\text{FHC}_1 \wedge \overline{\text{inner\_bad}}] + \sum_{k=2}^{\infty} \Pr[\text{FHC}_k \wedge \overline{\text{inner\_bad}}].$$

Note that, for  $b = 1$ , the event  $\text{FHC}_1$  is equivalent to the event `Univ`. On the other hand, for  $b = 2$ , the event  $\text{FHC}_1$  is equivalent to the event `Cross_Coll_Univ`. As we are bounding the probability of the joint event  $\text{FHC}_1$  and `inner_bad`, the probability of the event is zero when  $k = 1$ . Let  $k \geq 2$  and  $(i^{k+1}, b^k)$  be an alternating labeled path. Note that  $\lambda(i^{k+1}, b^k) = T_{i_2} + \dots + T_{i_k} + \mathbf{H}_{i_1}^2 + \mathbf{H}_{i_{k+1}}^2$ . Therefore, for  $k \geq 2$ , we have

$$\Pr[\text{FHC}_k \wedge \overline{\text{inner\_bad}}] \leq \Pr \left[ \exists i_1, \dots, i_{k+1} : \begin{array}{l} \mathbf{H}_{i_1}^1 = \mathbf{H}_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \\ \wedge \tau \text{ realizes } (i^{k+1}, b^k) \wedge \lambda(i^{k+1}, b^k) = 0^n \end{array} \right].$$

The number of tuples  $i^{k+1}$  such that  $\mathbf{H}_{i_1}^1 = \mathbf{H}_{i_{k+1}}^1$  and  $\lambda(i^{k+1}, b^k) = 0^n$  holds is at most  $3nq^{k-1}$ , as the number of pairs  $(i_1, i_{k+1})$  such that  $\mathbf{H}_{i_1}^1 = \mathbf{H}_{i_{k+1}}^1$  holds is at most  $3q$  (as we are considering `inner_bad` has not occurred) and for each choices of the pair  $(i_1, i_{k+1})$ , the number of ways  $i_2, \dots, i_k$  can be chosen such that  $T_{i_2} + \dots + T_{i_k} + \mathbf{H}_{i_1}^2 + \mathbf{H}_{i_{k+1}}^2 = 0^n$  holds is at most  $nq^{k-2}$ , where we use the fact that the number of multi-collision of  $T_i$  values is at most  $n$  (as we are considering `inner_bad` has not occurred). For each such choices of the tuple, the probability that  $\tau$  realizes  $(i^{k+1}, b^k)$  is boiled down to the probability of  $\mathbf{P}(\mathbf{H}_i^1)$  is a solution to  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$ , where  $\mathbf{P}$  is an  $n$ -bit uniform random permutation. Note that, the rank of  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  is exactly  $k - 1$  (as all hash values are distinct). Therefore, following Eqn. (2), we have

$$\Pr[\mathbb{E}_{\tau'}(i^{k+1}, b^k) \text{ holds}] \leq \frac{1}{(2^n - 1)_{k-1}}.$$

Now using the union bound we have

$$\Pr[\text{FHC}_k \wedge \overline{\text{inner\_bad}}] \leq \frac{3nq^{k-1}}{(2^n - 1)_{k-1}} \stackrel{(1)}{\leq} \frac{3n(2q)^{k-1}}{2^{n(k-1)}},$$

where (1) holds due to  $k \leq 2^{n-1}$ . Therefore, by assuming  $q \leq 2^{n-2}$ , we have

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q 3n \left( \frac{2q}{2^n} \right)^{k-1} \leq \frac{6nq}{2^n(1 - \frac{2q}{2^n})} \leq \frac{12nq}{2^n}. \quad (6)$$

**Lemma 9.** *Suppose FHC and inner\_bad do not occur. Then for any labeled query tree  $G$ , the system of equations  $\mathbb{E}_{\tau'}(G)$  are linearly independent. Hence,*

$$\Pr[\tau \text{ realizes } G \wedge \overline{\text{inner\_bad}} \wedge \overline{\text{FHC}}] \leq \frac{1}{(2^n - 1)_k}.$$

where  $k$  is the number of edges in  $G$ .

*Proof.* Suppose that the equations are not linearly independent, then there must exist distinct  $i, j$  from  $G$  such that  $H_i^1 = H_j^1$ . We choose  $i, j$  from  $G$  which has minimum length path between them. Let  $P$  be the path from  $i$  to  $j$ . Due to minimality, all hash values in the path do not collide with each other. In this path, we can skip all the nodes that appear with the same label in both neighboring edges. The reduced path has alternating labels and hence, FHC must hold (for this reduced path), which contradicts that FHC does not hold. Hence, the equations are linearly independent, which completes the first part of the proof.

Following Lemma 6,  $\tau$  realizes the labeled query graph  $G$  implies that  $P(H_i^1)$  is a solution to  $\mathbb{E}_{\tau'}(G)$ . Since  $\mathbb{E}_{\tau'}(G)$  is a system of linearly independent equations, the probability bound follows from Eqn. (2).  $\square$

TRANSCRIPT GRAPH. We fix an arbitrary transcript  $\tau = (M_1, T_1, H_1^1, H_1^2, X_1), \dots, (M_q, T_q, H_q^1, H_q^2, X_q)$ . We define a transcript graph  $G_\tau$  corresponding to  $\tau$  as follows:  $G_\tau = (V, \mathcal{E})$  is a bipartite graph that consists of a vertex set  $V_1 \sqcup V_2$  and an edge set  $\mathcal{E}$ .  $V_1$  consists of the variables  $X_i$  for all  $i \in [q]$  and  $V_2$  consists of the variables  $X_i \oplus T_i$  for all  $i \in [q]$ . We put a labeled edge between  $X_i$  and  $X_i \oplus T_i$  with label  $H_i^1$ . If two variable collides (e.g.,  $X_i = X_j$  or  $X_i \oplus T_i = X_j \oplus T_j$ ), then we merge the corresponding vertices together.

We construct a query graph  $\hat{G}_\tau$  corresponding to the fixed transcript  $\tau$ , where the vertices set is the set of query indices and we put an edge between two vertex  $i, j$  with label  $b$  if  $X_i = X_j$  (in that case the label is 1) or  $X_i \oplus T_i = X_j \oplus T_j$  (in that case the label is 2). Note that, if the constructed query graph contains a cycle  $C$  such that the label of each edge of the cycle is the same, then we break it by removing an arbitrary edge from the cycle. By definition, one can see that  $\hat{G}_\tau$  is realizable by the transcript  $\tau$ . In the following, we establish a relationship between the transcript graph  $G_\tau$  corresponding to a transcript  $\tau$  and the query graph  $\hat{G}_\tau$ .

**Lemma 10.** *Let  $\tau$  be a fixed transcript and  $G_\tau$  be the corresponding transcript graph.*

1. *If  $G_\tau$  contains an even length cycle, then  $\tau$  realizes an alternating labeled query cycle graph.*

2. Moreover, if  $G_\tau$  is a tree of size  $n$  (i.e., the number of vertices in  $G_\tau$ ), then  $\tau$  realizes a query tree  $\hat{G}_\tau$  having size exactly  $n - 1$ .

*Proof.* Let  $G_\tau$  contain an even-length cycle. Without loss of generality, let the cycle be  $C := (X_{i_1} = X_{i_2}, X_{i_2} \oplus T_{i_2} = X_{i_3} \oplus T_{i_3}, \dots, X_{i_k} \oplus T_{i_k} = X_{i_1} \oplus T_{i_1})$  of length  $k$ , where  $k$  is even. This set of  $k$  equalities generates the following alternating labeled query cycle graph  $\hat{G}_\tau: ((i_1, i_2, \dots, i_k, i_1), b^k)$ , where  $b_1 = 1$ . This completes the first part of the proof.

For the second part of the proof, the number of edges in  $G_\tau$  is  $n - 1$ . If all the edges in  $G_\tau$  are disjoint, then the total number of vertices would have been  $2(n - 1)$  as each edge gives rise to two vertices. But, according to the hypothesis, the graph  $G_\tau$  is connected and the number of vertices in  $G_\tau$  is  $n$ . Thus, there has been a merging of vertices. The number of such mergings is  $2(n - 1) - n = n - 2$ . Note that each merging occurs due to an equality relationship between  $X$  or  $X \oplus T$  values. Therefore, the number of such equalities is  $n - 2$ . Each such equality gives rise to an edge in a corresponding query graph. Hence, the total number of edges in the corresponding query graph realizable by  $\tau$  is  $n - 2$ . It is easy to see that as  $G_\tau$  is a tree, the corresponding query graph realizable by  $\tau$  is also a tree and thus, the number of vertices in the corresponding query graph is  $n - 1$ .  $\square$

**Definition 10 (Bad Attack Transcript).** We call an attainable attack transcript  $\tau$  to be **bad** if either its inner transcript  $\tau'$  is bad or first hash collision holds or one of the following conditions holds (also called mirror theory bad events):

- (Cycle): if  $G_\tau$  contains an even length cycle;
- (Component): if  $G_\tau$  contains a component with size at least  $n + 1$ ;
- (Zero\_Label\_Sum): if there exists an even length path in  $G_\tau$  such that the label sum of the path is zero;

Let  $X_{\text{id}}$  be the random variable of the transcript realized in the ideal world. Let  $\Theta_b$  denote the set of all bad transcripts. Then, the following lemma upper bounds the probability that a transcript is bad in the ideal world as follows:

**Lemma 11.** Let  $X_{\text{id}}$  and  $\Theta_b$  be defined as above. Then, we have

$$\Pr[X_{\text{id}} \in \Theta_b] \leq \frac{16nq}{2^n} + \left(\frac{4q}{2^n}\right)^n + \frac{12q}{2^n} + \left(\frac{2q}{2^n}\right)^n + \frac{q}{2^n} + \binom{q}{2} \left(\epsilon + \frac{\delta}{2^n}\right) + q\delta.$$

*Proof.* We have already obtained bounds for the probabilities of the events `inner_bad` and `FHC`. Now, we bound the probability of the subsequent bad events along with the event `FHC` and `inner_bad` have not occurred. As `FHC` and `inner_bad` does not occur, following Lemma 9 for any labeled graph  $G$ ,  $\mathbb{E}_{\tau'}(G)$  is a system of linearly independent equations. We use this fact to bound all the remaining bad events.

**A. Bounding (Cycle  $\wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}$ ):** Let  $G_\tau$  contains an even length cycle. Following Lemma 10,  $\tau$  realizes an alternating labeled query cycle graph  $G := (i^{k+1}, b^k) = ((i_1, i_2, \dots, i_k, i_{k+1}), b^k)$  (with  $i_{k+1} = i_1$ ), where  $b_1 = 1$ . Hence, the corresponding system of linear equations is

$$\mathbb{E}_{\tau'}(i^{k+1}, b^k) = \begin{cases} P(H_{i_1}^1) \oplus P(H_{i_2}^1) = H_{i_1}^2 \oplus H_{i_2}^2 \\ P(H_{i_2}^1) \oplus P(H_{i_3}^1) = H_{i_2}^2 \oplus H_{i_3}^2 \oplus T_{i_2} \oplus T_{i_3} \\ \vdots \\ P(H_{i_k}^1) \oplus P(H_{i_1}^1) = H_{i_k}^2 \oplus H_{i_1}^2 \oplus T_{i_k} \oplus T_{i_1} \end{cases}$$

Therefore, to bound the probability of the event that  $G_\tau$  contains an even length cycle, it is sufficient to bound the probability of the event that the system of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  holds. To ensure that  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  is consistent, we must have  $T_{i_1} \oplus T_{i_2} \oplus \dots \oplus T_{i_k} = 0^n$ . The number of tuples  $i^k := (i_1, \dots, i_k)$  satisfying  $T_{i_1} \oplus T_{i_2} \oplus \dots \oplus T_{i_k} = 0^n$  is at most  $nq^{k-1}$ , which follows from the multicollision of  $T_i$  values. Moreover, note that the rank of  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  is exactly the rank of  $\mathbb{E}_{\tau'}(i^k, b^{k-1})$  and the rank of the latter is exactly  $k-1$  as all the hash values  $H_{i_1}^1, \dots, H_{i_k}^1$  are distinct (follows from FHC does not hold). Therefore, by applying Lemma 9 for the labeled graph  $(i^k, (1, 2, \dots, 1))$ , we have

$$\Pr[\tau \text{ realizes } (i^k, (1, 2, \dots, 1)) \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{1}{(2^n - 1)_{k-1}}.$$

Thus, by assuming  $q \leq 2^{n-2}$  and  $k \leq 2^{n-1}$ , we have

$$\Pr[\text{Cycle} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q \frac{n(2q)^{k-1}}{2^{n(k-1)}} \leq \frac{4nq}{2^n}. \quad (7)$$

**B. Bounding (Component  $\wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}$ ):** Let  $G_\tau$  contains a component of size exactly  $n+1$ . For the purpose of the analysis, we concentrate on the largest component of  $G_\tau$  only. Let the component be  $C$  which is a tree of size  $n+1$ . Therefore, following Lemma 10,  $\tau$  realizes a labeled query tree  $G$  consisting of  $n$  vertices. Thus, it contains  $n-1$  edges. Each edge of the tree represents an equation of the form:

$$P(H_i^1) \oplus P(H_j^1) = \lambda(i, j, b), b \in \{1, 2\}.$$

Therefore, to bound the probability of the event that  $G_\tau$  contains a component of size at least  $n+1$ , it is sufficient to bound the probability of the event that  $\tau$  realizes a labeled query tree  $G$  of size  $n$ . Since, we consider the event  $\overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}$ , all the equations in the associated system of equations  $\mathbb{E}_{\tau'}(G)$  are linearly independent. Moreover, the number of indices is exactly  $n$ . Therefore, by applying Lemma 9 and by varying the choices for all possible indices, we have

$$\Pr[\text{Component} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{q^n}{(2^n - 1)_{n-1}} \leq q \left( \frac{2q}{2^n} \right)^{n-1} \leq \left( \frac{4q}{2^n} \right)^n. \quad (8)$$

**C. Bounding  $(\text{Zero\_Label\_Sum} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contain an even-length path of length  $k$  (i.e., the number of edges in the path), where  $k$  is even, such that the label sum of the path is zero. Following Lemma 10,  $\tau$  realizes an alternating labeled path  $(i^k, b^{k-1})$  of length  $k-1$ ,  $k \geq 2$ , where without loss of generality we assume that  $b_1 = 1$  (otherwise we consider the reverse ordering). Since  $\tau$  realizes  $(i^k, b^{k-1})$ , the following equalities

$$X_{i_1} = X_{i_2}, X_{i_2} \oplus T_{i_2} = X_{i_3} \oplus T_{i_3}, \dots, X_{i_{k-1}} = X_{i_k}$$

and the condition that the label sum of the path is zero holds, i.e.,  $H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = 0^n$  holds. Note that bounding the probability of the event that  $G_\tau$  contains a path of length  $k$  for any even  $k \geq 2$ , such that the label sum of the path is zero, it is sufficient to bound the probability of the event that  $\tau$  realizes an alternating labeled path  $(i^k, b^{k-1})$ . Let  $F$  denote the event  $\text{Zero\_Label\_Sum} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}$ . Thus, we have

$$\begin{aligned} \Pr[F] &= \Pr[\exists \text{ even } k : G_\tau \text{ contains path of length } k \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \\ &\leq \sum_{k=2}^{\infty} \Pr \left[ i_1, \dots, i_k : \begin{array}{l} \tau \text{ realizes } (i^k, b^{k-1}) \\ \wedge \bigoplus_{j=1}^k H_{i_j}^1 = 0^n \\ \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}} \end{array} \right]. \end{aligned}$$

The number of tuples  $i^k$  satisfying the condition  $H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = 0^n$  is at most  $3q^{k-1}$  (using the condition  $\overline{\text{Zero\_Hash\_Sum}}$ ). For a fixed such tuple, we bound

$$\Pr[\tau \text{ realizes } (i^k, b^{k-1}) \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}].$$

As  $\text{FHC}$  and  $\text{inner\_bad}$  does not hold, it follows that  $\mathbb{E}_{\tau'}(i^k, b^{k-1})$  is a system of linearly independent equations (by following Lemma 9), and thus we have

$$\Pr[\tau \text{ realizes } (i^k, b^{k-1}) \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{1}{(2^n - 1)_{k-1}}.$$

Thus, by assuming  $q \leq 2^{n-2}$  and  $k \leq 2^{n-1}$ , we have

$$\Pr[\text{Zero\_Label\_Sum} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^{\infty} \frac{3q^{k-1}}{(2^n - 1)_{k-1}} \leq 3 \sum_{k=2}^{\infty} \left( \frac{2q}{2^n} \right)^{k-1} \leq \frac{12q}{2^n} \quad (9)$$

Note that,

$$\begin{aligned} \Pr[X_{\text{id}} \in \Theta_b] &= \Pr[\text{Cycle} \vee \text{Component} \vee \text{Zero\_Label\_Sum} \vee \text{FHC} \vee \text{inner\_bad}] \\ &= \Pr[(\text{Cycle} \vee \text{Component} \vee \text{Zero\_Label\_Sum}) \wedge (\overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})] \\ &\quad + \Pr[\text{FHC} \vee \text{inner\_bad}] \\ &= \Pr[\text{Cycle} \wedge (\overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})] + \Pr[\text{Component} \wedge (\overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})] \\ &\quad + \Pr[\text{Zero\_Label\_Sum} \wedge (\overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})] + \Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \\ &\quad + \Pr[\text{inner\_bad}] \\ &\leq \frac{16nq}{2^n} + \left( \frac{4q}{2^n} \right)^n + \frac{12q}{2^n} + \left( \frac{2q}{2^n} \right)^n + \frac{q}{2^n} + \binom{q}{2} \left( \epsilon + \frac{\delta}{2^n} \right) + q\delta, \end{aligned}$$

where the last inequality follows from Lemma 7, Lemma 8, Eqn. (7), Eqn. (8), and Eqn. (9).  $\square$

## 4.2 Good Transcript Analysis

We fix an arbitrary good transcript  $\tau = ((M_1, T_1, H_1^1, H_1^2, X_1), \dots, (M_q, T_q, H_q^1, H_q^2, X_q))$ . Since  $\tau$  is a good transcript, the transcript graph  $G_\tau$  associated with  $\tau$  does not contain (a) any even length cycle, (b) the maximum size of its component is at most  $n$ , and (c) it does not contain any even length path such that the label sum of the path is zero. We now show that the probability of realizing  $\tau$  in the real world is as good as the probability of realizing it in the ideal world. In other words, we have the following result:

**Lemma 12.** *Let  $\tau = ((M_1, T_1, H_1^1, H_1^2, X_1), \dots, (M_q, T_q, H_q^1, H_q^2, X_q))$  be a good transcript. Then,*

$$\frac{\Pr[X_{\text{re}} = \tau]}{\Pr[X_{\text{id}} = \tau]} \geq 1.$$

*Proof.* We compute the ideal interpolation probability and the real interpolation probability for the transcript  $\tau$ . Computing the ideal interpolation probability is straightforward.  $T_i$ 's are distributed uniformly over  $\{0, 1\}^n$ , the hash key  $K$  is sampled uniformly at random, and finally the permutation  $P$  has been sampled uniformly at random from the set of all  $n$ -bit permutations. Let  $d$  be the number of distinct values of  $H_i^1$  and hence that of  $H_i^2 \oplus X_i$ . Thus, we have

$$\begin{aligned} \Pr[X_{\text{id}} = \tau] &= \Pr[\text{RF}(M_i) = T_i, \forall i \in [q]] \times \Pr[H_K(M_i) = (H_i^1, H_i^2) \forall i \in [q]] \\ &\quad \times \Pr[P(H_i^1) = X_i \oplus H_i^2, \forall i \in [q]] \\ &= \frac{1}{2^{nq}} \times \frac{1}{|\mathcal{K}|} \times \frac{1}{(2^n)_d} \end{aligned} \quad (10)$$

On the other hand, to compute a lower bound on the real interpolation probability, we need to compute the lower bound on the probability that the following system of equations holds, where  $P_0$  and  $P_1$  are two independent  $n$ -bit permutations:

$$\mathbb{E} = \begin{cases} P_0(X_1) \oplus P_1^{-1}(X_1 \oplus T_1) = H_1^1 \\ P_0(X_2) \oplus P_1^{-1}(X_2 \oplus T_2) = H_2^1 \\ \vdots \\ P_0(X_q) \oplus P_1^{-1}(X_q \oplus T_q) = H_q^1 \end{cases}$$

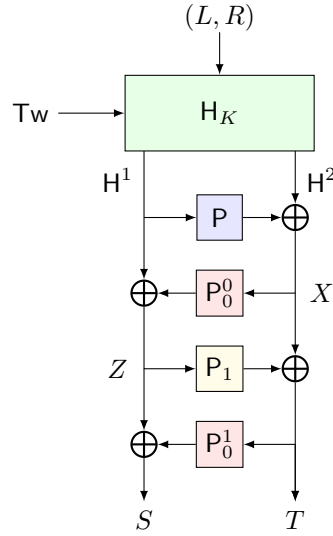
Since the transcript  $\tau$  is good, the graph  $G_\tau$  is acyclic, its component size is bounded by  $n$ , and for any even-length path, the label sum of the path is non-zero. Therefore, to compute the real interpolation probability, we have

$$\begin{aligned} \Pr[X_{\text{re}} = \tau] &= \Pr[H_K(M_i) = (H_i^1, H_i^2), \forall i \in [q]] \times \Pr[P(H_i^1) = X_i \oplus H_i^2, \forall i \in [q]] \\ &\quad \times \Pr[\mathbb{E} \text{ holds}] \\ &= \frac{1}{|\mathcal{K}|} \cdot \frac{1}{(2^n)_d} \cdot \frac{1}{2^{nq}}, \end{aligned} \quad (11)$$

where the last inequality follows from Lemma 2 as  $n^3 + n \leq 2^{n/2}$  for  $n \geq 29$  and  $qn^2 \leq 2^n/12$ . Therefore, by taking the ratio of Eqn. (11) to Eqn. (10), the result follows.

## 5 Hash-then-Four round Feistel: An Optimally Secure TPRP

In this section, we propose an optimally secure  $2n$ -bit TPRP, called  $\text{HF}^4$ , which is a composition of a  $2n$ -bit tweakable keyed hash function with a permutation-based four-round Feistel construction. It takes an arbitrary length tweak  $\text{Tw} \in \{0, 1\}^*$  and a  $2n$ -bit message  $(L, R)$ , which is fed through a  $2n$ -bit tweakable keyed hash function  $H_K$ . Let the output be  $(H^1, H^2)$ , which becomes the input of a four-round permutation-based Feistel construction. The construction is depicted in Fig. 5 and formally presented in Algorithm 2. The corresponding decryption algorithm is presented in 3.



**Figure 3.**  $\text{HF}^4$ : Hash-then-Four round Feistel Construction.  $H$  is a  $2n$ -bit tweakable keyed hash function,  $P$ ,  $P_0$ , and  $P_1$  are three independent  $n$ -bit permutations. We use the notation  $P_0^b(x)$  to denote  $P_0(\lfloor x \rfloor \| b)$ .



| Algorithm 2 $\text{HF}^4$ Encryption.                             | Algorithm 3 $\text{HF}^4$ Decryption.                             |
|-------------------------------------------------------------------|-------------------------------------------------------------------|
| 1: <b>function</b> $\text{HF}^4.\text{Enc}(K, \text{Tw}, (L, R))$ | 1: <b>function</b> $\text{HF}^4.\text{Dec}(K, \text{Tw}, (S, T))$ |
| 2: $(H^1, H^2) \leftarrow H_K(\text{Tw}, (L, R))$                 | 2: $Z \leftarrow P_0(\lfloor T \rfloor \  1) \oplus S$            |
| 3: $X \leftarrow P(H^1) \oplus H^2$                               | 3: $X \leftarrow P_1(Z) \oplus T$                                 |
| 4: $Z \leftarrow P_0(\lfloor X \rfloor \  0) \oplus H^1$          | 4: $H^1 \leftarrow P_0(\lfloor X \rfloor \  0) \oplus Z$          |
| 5: $T \leftarrow P_1(Z) \oplus X$                                 | 5: $H^2 \leftarrow P(H^1) \oplus X$                               |
| 6: $S \leftarrow P_0(\lfloor T \rfloor \  1) \oplus Y$            | 6: $(L, R) \leftarrow H_K^{-1}(\text{Tw}, (H^1, H^2))$            |
| 7: <b>return</b> $(S, T)$                                         | 7: <b>return</b> $(L, R)$                                         |

In the following, we show that  $\text{HF}^4$  achieves optimal tweakable PRP security. Formally, we have the following result.

**Theorem 3.** *Let  $H$  be an  $(\epsilon, \delta)$ - $2n$ -bit partial constant-sum-universal tweakable keyed hash function. Let  $P, P_0$ , and  $P_1$  be three independently sampled  $n$ -bit permutations. For any information theoretic deterministic distinguisher  $A$  that makes  $q$  queries to  $\text{HF}^4[H, P, P_0, P_1]$ , one has*

$$\begin{aligned} \text{Adv}_{\text{HF}^4[H, P, P_0, P_1]}^{\text{tprp}}(A) &\leq \frac{16nq}{2^n} + \frac{33q^2}{2^{2n}} + 2\left(\frac{4q}{2^n}\right)^n + \frac{8q^3\delta}{2^{2n}} + q(\delta + 2^{-n}) \\ &\quad + \binom{q}{2}(\epsilon + \delta \cdot 2^{-n}) \end{aligned}$$

assuming  $n \geq 35$  and  $qn^2 \leq 2^n/12$ .

*Proof.* Let  $A$  be a computationally unbounded deterministic distinguisher that interacts with the oracle  $\widetilde{\text{HF}}^4$  in the real world or with a  $2n$ -bit uniform tweakable random permutation  $\widetilde{\text{RP}}$  in the ideal world. This is known as the online phase. On the  $i$ th query adversary makes a query  $M_i := (\text{Tw}_i, (L_i, R_i)) \in \mathcal{T} \times \{0, 1\}^{2n}$  and obtain responses  $(S_i, T_i) \in \{0, 1\}^{2n}$ . Here,  $\mathcal{T}$  is the tweak space and  $\text{Tw}_i$ 's are the tweaks queried by the distinguisher. As  $A$  is deterministic, the randomness in the online interaction only comes from  $(S^q, T^q)$ . After the interaction, the oracle releases some additional information to the distinguisher (offline phase).

IDEAL WORLD OFFLINE PHASE. The ideal oracle samples a dummy hash key  $K$  and an  $n$ -bit random permutation  $P$  independently at random (and independent with  $(S^q, T^q)$ ). Then, it computes

$$(H_i^1, H_i^2) \leftarrow H_K(M_i), \quad X_i = P(H_i^1) \oplus H_i^2, \quad Y_i = X_i \oplus T_i, \quad i \in [q].$$

REAL WORLD OFFLINE PHASE. The real oracle computes  $(H_i^1, H_i^2, X_i, Y_i)_{i \in [q]}$  using the hash key  $K$  and the random permutation  $P$  which were sampled in the online phase.

ATTACK AND INNER TRANSCRIPT. In both worlds, the attack transcript used in the construction (the view of the distinguisher after the offline phase is over) is denoted as

$$\tau = ((M_1, (S_1, T_1), H_1^1, H_1^2, X_1, Y_1), \dots, (M_q, (S_q, T_q), H_q^1, H_q^2, X_q, Y_q)).$$

We write  $X'_i := \lfloor X_i \rfloor \| 0$  and  $T'_i := \lfloor T_i \rfloor \| 1$ . For any  $i, j \in [q]$ ,  $b \in \{1, 2\}$ , we define

$$\lambda(i, j, b) = \begin{cases} \lfloor H_i^2 \oplus H_j^2 \rfloor & \text{if } b = 1, \\ \lfloor H_i^2 \oplus H_j^2 \oplus T_i \oplus T_j \rfloor & \text{if } b = 2. \end{cases}$$

The *inner transcript* (a part of the attack transcript) is defined as

$$\tau' := ((M_1, (S_1, T_1), H_1^1, H_1^2), \dots, (M_q, (S_q, T_q), H_q^1, H_q^2)).$$

Note that  $\tau'$  is defined over only the tuple  $S^q, T^q$  and the hash key  $K$ .

ATTACK TRANSCRIPT IN THE REAL WORLD. Now we investigate the relationship among the values present in the attack transcript generated in the real world. We first note that

$$P_0(X'_i) \oplus P_1^{-1}(Y_i) = H_i^1, \quad P_0(T'_i) \oplus P_1^{-1}(Y_i) = S_i.$$

Note that  $X'_i = X'_j$  is equivalent in saying that

$$\lfloor P(H_i^1) \rfloor \oplus \lfloor P(H_j^1) \rfloor = \lambda(i, j, 1).$$

The equation  $Y_i = Y_j$  (after restricting to the most significant  $n - 1$  bits) implies that

$$\lfloor P(H_i^1) \rfloor \oplus \lfloor P(H_j^1) \rfloor = \lambda(i, j, 2).$$

SYSTEM OF LINEAR EQUATIONS. The above equations motivate us to consider a system of equations associated with an attack transcript of an ideal world. Let  $\{P_h : h \in \{0, 1\}^n\}$  be a collection of  $(n - 1)$ -bit variables (i.e., when we assign values to these variables, we assign only  $n - 1$  bit strings). Let  $G = (V \subseteq [q], E, B)$  be a  $\{1, 2, 3\}$ -labeled graph. For any edge  $e$  between  $i$  and  $j$  with a label  $b := B(e) \in \{1, 2\}$ , we associate an equation  $\mathbb{E}_{\tau'}(i, j, b)$  as follows:

$$P_{H_i^1} \oplus P_{H_j^1} = \lambda(i, j, b).$$

We call this graph a *query graph*. If it is a path or a cycle graph, then we call it *query path* or *query cycle* graph respectively. The collection of all equations over the set of all edges of  $G$  with label  $\{1, 2\}$  is denoted as  $\mathbb{E}_{\tau'}(G)$ .

**Definition 11.** Let  $i^{k+1}$  be an alternating tuple where each  $i_j \in [q]$ . Let  $G$  be a labeled walk over the sequence  $i^{k+1}$  with a label sequence  $b^k$  (i.e.,  $b_j$  is the label of the edge joined by  $i_j$  and  $i_{j+1}$ ). We also denote this labeled walk as  $(i^{k+1}, b^k)$ . We call the labeled walk alternating if  $b^k$  is an alternating sequence and  $b_k \neq b_1$  whenever  $i_{k+1} = i_1$  (i.e., when the walk is a loop).

Note that, the system of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  consists of the equations  $\mathbb{E}_{\tau'}(i_1, i_2, b_1)$ ,  $\mathbb{E}_{\tau'}(i_2, i_3, b_2)$ ,  $\dots$ ,  $\mathbb{E}_{\tau'}(i_k, i_{k+1}, b_k)$ .

**Definition 12.** Let  $G$  be a labeled graph. We say that  $G$  is realized by an attack transcript  $\tau$  if for all edge  $\{i, j\}$  in  $G$  with a label  $b$ , we have

$$\begin{cases} X'_i = X'_j & \text{if } b = 1 \\ X_i \oplus T_i = X_j \oplus T_j & \text{if } b = 2 \\ T'_i = T'_j & \text{if } b = 3. \end{cases}$$

For a graph  $G'$  with label 3 edges only, we call  $T^q$  realizes  $G'$  if for all edges between  $i, j$  in  $G'$ , we have  $T'_i = T'_j$ .

When the label is 2, we lose one bit of equality, and we may consider  $X'_i \oplus T'_i = X'_j \oplus T'_j$ . This would help to write all the equations in terms of  $X'_i$  and  $T'_i$  variables. Let  $G$  be a labeled graph over a vertex set  $V \subseteq [q]$  with label set  $\{1, 2, 3\}$ . We write  $G_3$  (resp.  $G_{1,2}$ ) to denote the subgraph of  $G$  consisting of all label 3 edges (resp. label 1 and 2 edges).

**Lemma 13.** Let  $G$  be a labeled graph over a vertex set  $V \subseteq [q]$  with label set  $\{1, 2, 3\}$ . If  $\tau$  realizes a labeled graph  $G = (V, E, B)$  then

1.  $T^q$  realizes  $G_3$ ;
2.  $(P_h = \lfloor P(h) \rfloor, \forall h \in \{0, 1\}^n)$  is a solution for the system  $\mathbb{E}_{\tau'}(G_{1,2})$ .

FIRST HASH COLLISION (FHC) AND FIRST DEPENDENCY (FD). Let us fix a transcript  $\tau$ . For any  $k \geq 1$ , we define the following set based on its inner transcript:

$$\mathcal{I}^{(k)} := \{i^{k+1} \in [q]^{k+1} \mid i_1, \dots, i_k, H_{i_1}^1, \dots, H_{i_k}^1 \text{ are distinct, } H_{i_1}^1 = H_{i_{k+1}}^1\}.$$

Note that when  $i_{k+1} = i_1$ , the hash equality condition is trivially satisfied. In this case, we are actually considering a cycle where all hash values for each query index are distinct.

**Definition 13 (FHC and FD Bad Event).** We say that the first hash collision (FHC) event holds if there exists an  $k \geq 1$  such that  $i^{k+1} \in \mathcal{I}^{(k)}$  and the attack transcript  $\tau$  realizes an alternating labeled path  $(i^{k+1}, t^k := (t_1, \dots, t_k))$  with  $t_j = 3$  for some  $j \in [k]$ .

Moreover, we say that the first dependency (FD) event holds if there exists an  $k \geq 1$  such that  $i^{k+1} \in \mathcal{I}^{(k)}$  and the attack transcript  $\tau$  realizes an alternating labeled path  $(i^{k+1}, t^k := (t_1, \dots, t_k))$  with  $t_j \neq 3$  for all  $j \in [k]$ , and the label sum of the path is zero, i.e.,

$$\lambda(i^{k+1}, t^k) := \bigoplus_{j=1}^k \lambda(i_j, i_{j+1}, t_j) = 0^{n-1}.$$

**Definition 14.** We call an inner transcript  $\tau'$  to be **bad** if one of the following conditions holds:

- **Multi-Coll** :  $\max_a |\{i : T_i = a\}| \geq n$  or  $\max_a |\{i : S_i = a\}| \geq n$ ;
- **Coll** :  $|\{(i, j) : T_i = T_j, i \neq j\}| \geq q$ ;
- **Univ** :  $\exists i \neq j \in [q] : H_K(M_i) = H_K(M_j)$ ;
- **Cross\_Coll\_Univ** :  $\exists i, j \in [q]$  such that  $H_i^1 = H_j^1 \wedge H_i^2 \oplus T_i = H_j^2 \oplus T_j$ .
- **Constant\_Hash\_Sum** : For some  $k \geq 2$ , and for any  $a \in \{0, 1\}^n$

$$\left| \left\{ (i_1, i_2, \dots, i_k) \in ([q])_k : \bigoplus_{j \in I} H_j^1 = a \right\} \right| \geq 3q^{k-1}.$$

Let us define an event **inner\_bad** as the union of the above events, i.e.,

$$\text{inner\_bad} := \left( \text{Multi-Coll} \vee \text{Coll} \vee \text{Univ} \vee \text{Cross\_Coll\_Univ} \vee \text{Constant\_Hash\_Sum} \right).$$

**Lemma 14.**

$$\Pr[\text{inner\_bad}] \leq \left( \frac{4q}{2^n} \right)^n + q(\delta_{\text{csu}} + 2^{-n}) + \left( \frac{q}{2} \right) (\delta_{\text{univ}} + \delta_{\text{csu}} \cdot 2^{-n}).$$

*Proof.* We individually upper bound the probability of the event, and then we combine them by virtue of the union bound to yield the result.

**Bounding Multi-Coll:** For a fixed choices of indices  $i_1, i_2, \dots, i_n$ , the event  $T_{i_1} = T_{i_2} = \dots = T_{i_n}$  holds with probability  $2^{-n(n-1)}$ , as there are  $n-1$  equalities and each such equality contributes  $2^{-n}$  to the probability of the event. Moreover, the number of possible choices of indices is at most  $q^n$ . Similarly, for a fixed choices of indices  $i_1, i_2, \dots, i_n$ , the event  $S_{i_1} = S_{i_2} = \dots = S_{i_n}$  holds with probability  $2^{-n(n-1)}$ , as there are  $n-1$  equalities and each such equality contributes  $2^{-n}$  to the probability of the event. Thus, we have

$$\Pr[\text{Multi-Coll}] \leq \frac{2q^n}{2^{n(n-1)}} \leq \left( \frac{4q}{2^n} \right)^n. \quad (12)$$

**Bounding Coll:** Let us consider a pair of indices  $(i, j)$ . We call this pair to be colliding if  $T_i = T_j$ . Note that the expected number of colliding pair is at most  $q^2 \cdot 2^{-n}$  as for a fixed pair of indices, the probability of the event  $T_i = T_j$  holds is at most  $2^{-n}$ . Therefore, by applying Markov inequality, we obtain

$$\Pr[\text{Coll}] \leq \frac{q}{2^n}.$$

**Bounding Univ and Cross\_Coll\_Univ:** For a fixed choices of indices  $i \neq j$ , the event  $H_K(M_i) = H_K(M_j)$  holds with probability  $\delta_{\text{univ}}$ . By varying over all possible choices of indices  $i \neq j$ , we have

$$\Pr[\text{Univ}] \leq \left( \frac{q}{2} \right) \epsilon. \quad (13)$$

To bound the probability of the `Cross_Coll_Univ` event, let us assume  $i < j$ . For a fixed choices of indices  $i < j$ , we bound the probability of the event  $H_i^1 = H_j^1$  and  $H_i^2 \oplus T_i = H_j^2 \oplus T_j$ . By conditioning on  $T_i$  and  $M_j$ , the event  $H_i^1 = H_j^1$  holds with probability  $\delta$  (due to the universal property of the  $H^1$  function). Once we fix the hash key,  $H_i^2 \oplus H_j^2$  gets also fixed. Thus, the event  $T_j = T_i \oplus H_i^2 \oplus H_j^2$  holds with probability  $2^{-n}$  (due to the randomness of  $T_j$ ). Note that we already have conditioned  $T_i$  and hence we cannot use its entropy to bound the event. By varying over all possible choices of indices  $i \neq j$ , we have

$$\Pr[\text{Cross\_Coll\_Univ}] \leq \binom{q}{2} \delta \cdot 2^{-n}. \quad (14)$$

**Bounding Constant\_Hash\_Sum:** Let  $X$  denotes the cardinality of the set  $\{(i_1, i_2, \dots, i_k) : H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = a\}$ . For each tuple  $(i_1, \dots, i_k) \in ([q])_k$ , we define an indicator random variable  $X_{i_1, \dots, i_k}$  as follows: it takes the value 1 if  $H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = a$  and 0 otherwise. Then, we have

$$X = \sum_{i_1, \dots, i_k} X_{i_1, \dots, i_k}.$$

Therefore, by linearity of expectations, we have

$$\mathbf{Exp}[X] = (q)_k \Pr[H_{i_1}^1 \oplus \dots \oplus H_{i_k}^1 = a] \leq q^k \delta.$$

Therefore, by applying Markov's inequality, we have

$$\Pr[\text{Constant\_Hash\_Sum}] = \Pr[X > 3q^{k-1}] \leq \frac{\mathbf{Exp}[X]}{3q^{k-1}} \leq q\delta.$$

Note that the event `Constant_Hash_Sum` for  $k = 2$  and  $a = 0^n$  is the same as saying that the number of colliding pairs of  $H^1$  values is at least  $3q$ .  $\square$

**Lemma 15.**  $\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \frac{8q^3 \delta_{\text{csu}}}{2^{2n}}.$

*Proof.* For any  $k \geq 1$ , we define the event  $\text{FHC}_k$  as follows: we say that  $\text{FHC}_k$  holds if (i)  $i^{k+1} \in \mathcal{I}^{(k)}$ , and (ii)  $\tau$  realizes an alternating label path  $(i^{k+1}, t^k)$ . Thus,

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \Pr[\text{FHC}_1 \wedge \overline{\text{inner\_bad}}] + \sum_{k=2}^{\infty} \Pr[\text{FHC}_k \wedge \overline{\text{inner\_bad}}].$$

Note that the event  $\text{FHC}_1$  is equivalent to the event `Univ` or `Cross_Coll_Univ`. As we are bounding the probability of the joint event  $\text{FHC}_1$  and  $\overline{\text{inner\_bad}}$ , the probability of the event is zero when  $k = 1$ . Let  $k \geq 2$  and  $(i^{k+1} := (i_1, \dots, i_{k+1}), t^k)$  be an alternating labeled path. Therefore, for  $k \geq 2$ , we have

$$\Pr[\text{FHC}_k \wedge \overline{\text{inner\_bad}}] \leq \Pr \left[ \exists i_1, \dots, i_{k+1} : \begin{array}{l} H_{i_1}^1 = H_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \\ \wedge \tau \text{ realizes } (i^{k+1}, t^k) \end{array} \right].$$

Let  $G = (i^{k+1}, t^k)$  be a fixed alternating labeled path. Let there be  $r \geq 1$  edges in the path with label 3. Let  $G_{1,2}$  be the subgraph of  $G$  by removing all label 3 edges from  $G$ , and  $G_3$  be the collection of all edges of  $G$  with label 3. Thus, the probability that  $\tau$  realizes the alternating labeled path  $G = (i^{k+1}, t^k)$  is boiled down to the probability of  $P(H_i^1)$  is a solution to  $\mathbb{E}_{\tau'}(G_{1,2})$  and  $T_i = T_j$  for all  $\{i, j\} \in E(G_3)$ . Therefore, by applying the union bound, we have

$$\Pr \left[ \exists i_1, \dots, i_{k+1} : \begin{array}{l} H_{i_1}^1 = H_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \\ \wedge \tau \text{ realizes } (i^{k+1}, t^k) \end{array} \right] \leq \sum_{i_1, \dots, i_k} \Pr \left[ \begin{array}{l} H_{i_1}^1 = H_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \\ \wedge \tau \text{ realizes } (i^{k+1}, t^k) \end{array} \right]$$

Let  $E$  denote the event  $H_{i_1}^1 = H_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \wedge \tau \text{ realizes } (i^{k+1}, t^k)$ .

$$\begin{aligned} \Pr[E] &= \Pr \left[ \begin{array}{l} H_{i_1}^1 = H_{i_{k+1}}^1 \wedge T_{i_j} = T_{i_{j+1}} \forall \{i_j, i_{j+1}\} \in E(G_3) \wedge \mathbb{E}_{\tau'}(G_{1,2}) \text{ holds} \wedge \overline{\text{inner\_bad}} \end{array} \right] \\ &= \Pr \left[ \underbrace{T_{i_j} = T_{i_{j+1}}, \forall \{i_j, i_{j+1}\} \in E(G_3) \wedge \overline{\text{inner\_bad}}}_{E_1} \right] \times \Pr \left[ \underbrace{H_{i_1}^1 = H_{i_{k+1}}^1}_{E_2} \mid E_1 \right] \\ &\quad \times \Pr \left[ \underbrace{\mathbb{E}_{\tau'}(G_{1,2}) \text{ holds}}_{E_3} \mid E_2, E_1 \right] \end{aligned}$$

Since  $G_3$  is a collection of  $r$  many label 3 edges,  $\Pr[E_1] = 2^{-nr}$ . Moreover, we have  $\Pr[E_2] \leq \delta_{\text{csu}}$ . Finally,  $G_{1,2}$  has  $k - r$  many edges and the rank of  $\mathbb{E}_{\tau'}(G_{1,2})$  is exactly  $k - r$ . Therefore, by following Eqn. (2), we have

$$\Pr[E_3] \leq \frac{1}{(2^n)_{k-r}}.$$

Therefore, by varying over all possible choices of indices, we have

$$\Pr \left[ \exists i_1, \dots, i_{k+1} : \begin{array}{l} H_{i_1}^1 = H_{i_{k+1}}^1 \wedge \overline{\text{inner\_bad}} \\ \wedge \tau \text{ realizes } (i^{k+1}, t^k) \end{array} \right] \leq \frac{2^k q^{k+1} \delta}{2^{nk}}.$$

Hence, we have

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q \frac{2^k q^{k+1} \delta}{2^{nk}} \leq q \delta \sum_{k=2}^{\infty} \left( \frac{2q}{2^n} \right)^k \leq \frac{8q^3 \delta}{2^{2n}}.$$

This completes the proof.  $\square$

**Lemma 16.**  $\Pr[\text{FD} \wedge \overline{\text{inner\_bad}}] \leq \frac{12nq}{2^n}.$

Proof of this lemma is identical to that of Lemma 8 and hence we omit it. Now, we define an event  $\text{EFHC} := \text{FHC} \vee \text{FD}$  and following Lemma 15 and Lemma 16 we have

$$\Pr[\text{EFHC} \wedge \overline{\text{inner\_bad}}] \leq \frac{8q^3 \delta}{2^{2n}} + \frac{12nq}{2^n}. \quad (15)$$

**Lemma 17.** *Let EFHC and inner\_bad do not occur. Then for any labeled tree  $G = (V, E, B)$ , the system of equations  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent. Hence,*

$$p := \Pr[\tau \text{ realizes } G \wedge \overline{\text{inner\_bad}} \wedge \overline{\text{EFHC}}] \leq \frac{1}{(2^n - 1)_{k-r}} \cdot \frac{1}{2^{nr}}.$$

where  $r = |E \setminus E(G_{1,2})|$  and  $k$  is the total number of edges in  $G$ .

*Proof.* Suppose that the equations are not linearly independent, then there must exist distinct  $i, j$  from  $G$  such that  $H_i^1 = H_j^1$ . We choose  $i, j$  from  $G$  that has a minimum length path between them. Let  $P$  be the path from  $i$  to  $j$ . Due to minimality, all hash values in the paths do not collide among themselves. In this path, we can skip all the nodes that appear with the same label in both neighboring edges. The reduced path has alternating labels with  $\{1, 2, 3\}$ . Now, if there are no edges with label 3 in the reduced path, then FD must hold (for this reduced path). On the other hand, if there is an edge with the label 3 on the reduced path, then FHC must hold (for this reduced path). So, for both cases, we will reach a contradiction that FHC, FD do not occur. Hence, the system of equations  $\mathbb{E}_{\tau'}(G)$  and thus  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent. This completes the first part of the proof.

Let  $\text{Bad} = \text{inner\_bad} \vee \text{EFHC}$  and we have shown that the equations  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent. Following the definition of a transcript realizing a labeled graph  $G$ , we have

$$\begin{aligned} p &\leq \Pr[\mathbf{P}(h) \text{ is a solution to } \mathbb{E}_{\tau'}(G_{1,2}) \wedge \overline{\text{Bad}} \wedge T_i = T_j, \{i, j\} \in E(G_{1,2})] \\ &\leq \Pr[\mathbf{P}(h) \text{ is a solution to } \mathbb{E}_{\tau'}(G_{1,2}) \wedge \overline{\text{Bad}} \mid T_i = T_j, \{i, j\} \in E(G_{1,2})] \\ &\quad \cdot \Pr[T_i = T_j, \{i, j\} \in E(G_{1,2})] \\ &\leq \frac{1}{(2^n - 1)_{k-r}} \cdot \frac{1}{2^{nr}}, \end{aligned}$$

which concludes the proof.  $\square$

**Lemma 18.** *Let EFHC and inner\_bad do not occur. Then for any alternating labeled cycle  $G = (i^{k+1} = (i_1, \dots, i_{k+1}), t^k)$  such that  $t_j = 3$  for some  $j$ , the system of equations  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent, where  $G_{1,2}$  is the subgraph of  $G$  by removing all its label 3 edges. Hence,*

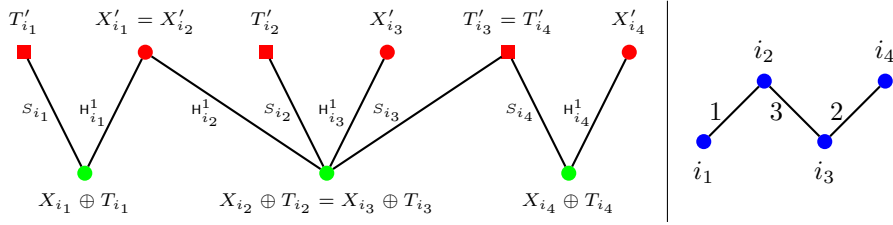
$$p := \Pr[\tau \text{ realizes } G \wedge \overline{\text{inner\_bad}} \wedge \overline{\text{EFHC}}] \leq \frac{1}{(2^n - 1)_{k-r}} \cdot \frac{1}{2^{nr}}.$$

where  $r = |E \setminus E(G_{1,2})|$  and  $k$  is the total number of edges in  $G$ .

*Proof.* Suppose that the equations are not linearly independent, then there must exist distinct  $i, j$  from  $G$  such that  $H_i^1 = H_j^1$ . We choose  $i, j$  from  $G$  that has a minimum length path between them. Let  $P$  be the path from  $i$  to  $j$ . Due to minimality, all hash values in the paths do not collide among themselves. This path has alternating labels with  $\{1, 2, 3\}$ . Thus, FHC must hold for this path.

So, we reach a contradiction that EFHC does not occur. Hence, the system of equations  $\mathbb{E}_{\tau'}(G)$  and thus  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent. This completes the first part of the proof. The second part of the proof is identical to that of Lemma 17 and hence we omit it.  $\square$

**TRANSCRIPT GRAPH.** We fix an arbitrary transcript  $\tau = (M_1, (S_1, T_1), H_1^1, H_1^2, X_1, Y_1), \dots, (M_q, (S_q, T_q), H_q^1, H_q^2, X_q, Y_q)$ . We define a transcript graph  $G_\tau$  corresponding to  $\tau$  as follows:  $G_\tau = (V, \mathcal{E})$  is a bipartite graph that consists of a vertex set  $V_1 \sqcup V_2$  and an edge set  $\mathcal{E}$ .  $V_1$  consists of the variables  $X'_i \cup T'_i$  for all  $i \in [q]$  and  $V_2$  consists of the variables  $X_i \oplus T_i$  for all  $i \in [q]$ . For each  $i \in [q]$ , we put a labeled edge between  $X'_i$  and  $X_i \oplus T_i$  with label  $H_i^1$  and another labeled edge between  $T'_i$  and  $X_i \oplus T_i$  with label  $S_i$ . If two variable collides (e.g.,  $X'_i = X'_j$  or  $X_i \oplus T_i = X_j \oplus T_j$  or  $T'_i = T'_j$ ), then we merge the corresponding vertices together.



**Figure 4.** A Transcript Graph (left) and its corresponding Query Graph (right)

We construct a query graph  $\hat{G}_\tau$  corresponding to the fixed transcript  $\tau$ , where the vertices set is the set of query indices and we put an edge between two vertex  $i, j$  with label  $b$  if  $X'_i = X'_j$  (in that case the label is 1) or  $X_i \oplus T_i = X_j \oplus T_j$  (in that case the label is 2) or  $T'_i = T'_j$  (in that case the label is 3). Note that if the constructed query graph contains a cycle  $C$  such that the label of each edge of the cycle is the same, then we break it by removing an arbitrary edge from the cycle. By definition, one can see that  $\hat{G}_\tau$  is realizable by the transcript  $\tau$ . In the following, we establish a relationship between the transcript graph corresponding to a transcript  $\tau$  and the query graph  $\hat{G}_\tau$ .

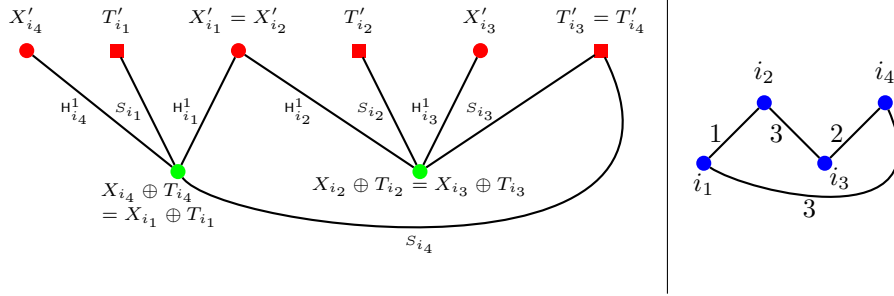
**Lemma 19.** *Let  $\tau$  be a fixed transcript and  $G_\tau$  be the corresponding transcript graph.*

1. *If  $G_\tau$  contains an even length cycle, then  $\tau$  realizes an alternating labeled query cycle graph.*
2. *Moreover, if  $G_\tau$  is a tree of size  $2k + 1$  (i.e., the number of vertices in  $G_\tau$ ), then  $\tau$  realizes a query tree  $\hat{G}_\tau$  having size exactly  $k$ .*

*Proof.* Let  $G_\tau$  contain an even-length cycle. Then, by following the argument of Lemma 10, each edge of the cycle gives rise to an equality, which in turn



generates an alternating labeled query cycle graph  $\hat{G}_\tau: ((i_1, i_2, \dots, i_k, i_1), b^k)$ . For example, following Fig. 5, one can see that a cycle of length four in the transcript graph yields a cycle in the associated query graph. This completes the first part of the proof.



**Figure 5.**  $G_\tau$  contains a cycle (left) and its corresponding Query Cycle Graph (right)

For the second part of the proof, the number of edges in  $G_\tau$  is  $2k$ . If all the edges in  $G_\tau$  are disjoint, then the total number of vertices would have been  $3k$  as two edge gives rise to three distinct vertices. But, as the number of vertices in  $G_\tau$  is  $2k + 1$ , there has been a merging of vertices. The number of such merging is  $3k - (2k + 1) = k - 1$ . Note that each merging occurs due to an equality relationship between  $X'$  or  $X \oplus T$ , or  $T'$  values. Therefore, the number of such equalities is  $k - 1$ . Each such equality gives rise to an edge in a query graph. Hence, the number of edges in the corresponding query graph realizable by  $\tau$  is  $k - 1$ . For example, one can see from Fig. 5 that the component size in the transcript graph is nine, which means there would have been 12 distinct vertices, but there are nine distinct vertices in the graph, which implies that there are three equalities. Hence, it gives rise to a query graph with three edges and four vertices. It is easy to see that as  $G_\tau$  is a tree, the corresponding query graph realizable by  $\tau$  is also a tree and thus, the number of vertices in the corresponding query graph is  $k$ .  $\square$

**Definition 15 (Bad Attack Transcript).** We call an attainable transcript  $\tau$  to be **bad** if either its inner transcript  $\tau'$  is bad, or first hash collision holds, or first dependency holds, or one of the following conditions holds:

- (Cycle): if  $G_\tau$  contains an even length cycle.
- (Component): if  $G_\tau$  contains a component with size at least  $2n + 1$ .
- (Zero\_Label\_Sum): if there exists an even length path in  $G_\tau$  such that the label sum of the path is zero;

Let  $X_{\text{id}}$  be the random variable of the transcript realized in the ideal world. Let  $\Theta_b$  denote the set of all bad transcripts. Then, the following lemma upper bounds the probability that a transcript is bad in the ideal world as follows:

**Lemma 20.** *Let  $X_{\text{id}}$  and  $\Theta_b$  be defined as above. Then, we have*

$$\Pr[X_{\text{id}} \in \Theta_b] \leq \frac{16nq}{2^n} + \frac{32q^2}{2^{2n}} + 2\left(\frac{4q}{2^n}\right)^n + \frac{8q^3\delta}{2^{2n}} + q(\delta + 2^{-n}) + \binom{q}{2}(\epsilon + \delta \cdot 2^{-n}).$$

*Proof.* We have already obtained bounds for the probabilities that an inner transcript is bad. Now, we bound the probability of the subsequent bad events (mirror theory bad events) along with the event  $\overline{\text{EFHC}}$  and  $\text{inner\_bad}$  (for the inner transcript) have not occurred. As  $\overline{\text{EFHC}}$  and  $\text{inner\_bad}$  does not occur, following Lemma 17 and Lemma 18, for any labeled graph  $G$ ,  $\mathbb{E}_{\tau'}(G_{1,2})$  is a system of linearly independent equations, where  $G_{1,2}$  is a labeled subgraph of  $G$  by removing all its label 3 edges. We use this fact to bound all the remaining bad events.

**A. Bounding  $(\text{Cycle} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contain an even length cycle. Following Lemma 19,  $\tau$  realizes an alternating labeled query cycle graph  $\hat{G}_\tau$ . Therefore, to bound the probability of the event that there exists some even  $k$  such that  $G_\tau$  contains a cycle of length  $k$ , it is sufficient to bound the probability of the event that  $\tau$  realizes an alternating labeled query cycle graph  $\hat{G}_\tau$ . In other words,

$$\begin{aligned} \Pr[\text{Cycle} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] &\leq \Pr\left[\exists k : \begin{array}{l} \tau \text{ realizes alternating labeled} \\ \text{query cycle graph } (i^{k+1}, t^k) \text{ of length } k \end{array}\right] \\ &\leq \sum_{k=2}^{\infty} \Pr\left[\exists k : \begin{array}{l} \tau \text{ realizes alternating labeled} \\ \text{query cycle graph } (i^{k+1}, t^k) \text{ of length } k \end{array}\right] \end{aligned}$$

Thus, we bound the probability of this latter event in two cases: (a) when the cycle contains no label 3 edges and (b) when the cycle contains at least one label 3 edge.

**Case-(a):** Suppose  $\tau$  realizes an alternating labeled query cycle graph  $(i^{k+1}, t^k)$  with  $i_{k+1} = i_1, k \geq 2$ . Let us assume that  $t_j \neq 3$  for all  $j \in [k]$ , and without loss of generality, we assume that  $t_1 = 1$ . Since  $\tau$  realizes  $(i^{k+1}, t^k)$ , the following equalities hold

$$X_{i_1} = X_{i_2}, X_{i_2} \oplus T_{i_2} = X_{i_3} \oplus T_{i_3}, \dots, X_{i_k} \oplus T_{i_k} = X_{i_1} \oplus T_{i_1}.$$

and hence the corresponding system of linear equation is

$$\mathbb{E}_{\tau'}((i^{k+1}, t^k)) = \begin{cases} P(H_{i_1}^1) \oplus P(H_{i_2}^1) = H_{i_1}^2 \oplus H_{i_2}^2 \\ P(H_{i_2}^1) \oplus P(H_{i_3}^1) = H_{i_2}^2 \oplus H_{i_3}^2 \oplus T_{i_2} \oplus T_{i_3} \\ \vdots \\ P(H_{i_k}^1) \oplus P(H_{i_1}^1) = H_{i_k}^2 \oplus H_{i_1}^2 \oplus T_{i_k} \oplus T_{i_1} \end{cases}$$

To ensure that  $\mathbb{E}_{\tau'}(i^{k+1}, t^k)$  is consistent, we must have  $T_{i_1} \oplus T_{i_2} \oplus \dots \oplus T_{i_k} = 0^n$ . The number of tuples  $i^k := (i_1, \dots, i_k)$  satisfying  $T_{i_1} \oplus T_{i_2} \oplus \dots \oplus T_{i_k} = 0^n$  is at most  $nq^{k-1}$ , which follows from the multicollision of  $T_i$  values. Moreover, note that the rank of  $\mathbb{E}_{\tau'}(i^{k+1}, t^k)$  is exactly the rank of  $\mathbb{E}_{\tau'}(i^k, t^{k-1})$  and the rank of the latter is exactly  $k-1$  as all the hash values  $H_{i_1}^1, \dots, H_{i_k}^1$  are distinct (follows from FHC and FD does not hold). Therefore, by applying Lemma 17 with  $r = 0$  for the labeled graph  $(i^k, (1, 2, \dots, 1))$ , we have

$$\Pr[\tau \text{ realizes } \mathbb{E}_{\tau'}(i^k, (1, 2, \dots, 1)) \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{1}{(2^n - 1)_{k-1}}.$$

Thus, by assuming  $q \leq 2^{n-2}$  and  $k \leq 2^{n-1}$ , we have

$$\Pr[\text{Cycle} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q \frac{n(2q)^{k-1}}{2^{n(k-1)}} \leq \frac{4nq}{2^n}. \quad (16)$$

**Case-(b):** On the other hand, suppose  $\tau$  realizes an alternating labeled query cycle graph  $(i^{k+1}, t^k)$  with  $i_{k+1} = i_1, k \geq 2$ , with  $t_j = 3$  for some  $j \in [k]$ . Without loss of generality, we assume that  $t_1 = 1$ . Since  $\tau$  realizes  $(i^{k+1}, t^k)$ , by applying Lemma 18, we have

$$\Pr[\tau \text{ realizes } \mathbb{E}_{\tau'}(i^{k+1}, t^k) \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{1}{(2^n - 1)_{k-r}} \cdot \frac{1}{2^{nr}},$$

where  $r$  is number of label 3 edges. Thus, by assuming  $q \leq 2^{n-2}$  and  $k \leq 2^{n-1}$  and by varying over all possible choices of indices, we have

$$\Pr[\text{Cycle} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q \frac{2^{k-r} q^k}{2^{nk}} \leq \sum_{k=2}^q \left(\frac{2q}{2^n}\right)^k \leq \frac{8q^2}{2^{2n}}. \quad (17)$$

By combining the above two cases, we have

$$\Pr[\text{Cycle} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{4nq}{2^n} + \frac{8q^2}{2^{2n}} \quad (18)$$

**B. Bounding  $(\text{Component} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contain a component of size exactly  $2n + 1$ . For the purpose of the analysis, we concentrate on the largest component of  $G_\tau$  only. Let the component be  $C$ , which is a tree of size  $2n + 1$ . Therefore, following Lemma 19,  $\tau$  realizes a labeled query tree  $\hat{G}_\tau$  consisting of  $n$  vertices. Thus, it contains  $n - 1$  edges. Therefore, to bound the probability of the event that  $G_\tau$  contains a component of size at least  $2n + 1$ , it is sufficient to bound the probability of the event that  $\tau$  realizes a labeled query tree  $\hat{G}_\tau$  of size  $n$ . Since, we consider the event  $\overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}$ , following Lemma 17, all the equations in the associated system of equations  $\mathbb{E}_{\tau'}(G_{1,2})$  are linearly independent, where  $G_{1,2}$  is the subgraph of  $G$  by removing all its label 3 edges. Therefore, by applying Lemma 17 and by varying the choices for all possible indices, we have

$$\Pr[\text{Component} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{q^n}{(2^n - 1)_{n-1-r}} \cdot \frac{1}{2^{nr}} \leq q \left(\frac{2q}{2^n}\right)^n \leq \left(\frac{4q}{2^n}\right)^n \quad (19)$$

**C. Bounding  $(\text{Zero\_Label\_Sum} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contains an even length path  $P$  of length  $k$  (i.e., the number of edges in  $P$ ),  $k$  is even, such that the label sum of the path  $P$  is zero. Following Lemma 19,  $\tau$  realizes an alternating labeled query path  $P'$  and the condition that the label sum of the path  $P$  is zero holds, i.e.,

$$\bigoplus_{j \in I_1} H_j^1 \oplus \bigoplus_{j \in I_2} S_j = 0^n,$$

for some sets  $I_1$  and  $I_2$  with  $I_1 \cup I_2 = \{i_1, \dots, i_{k'+1}\}$ . Note that, bounding the probability of the event that  $G_\tau$  contains a path of length  $k$  for any even  $k \geq 2$ , such that the label sum of the path is zero, it is sufficient to bound the probability of the event that

$$(\exists k' \geq 2), i_1, \dots, i_{k'+1} : \tau \text{ realizes } (i^{k'+1}, t^{k'}), \bigoplus_{j \in I_1} H_j^1 \oplus \bigoplus_{j \in I_2} S_j = 0^n.$$

Let  $F$  denote the event  $\text{Zero\_Label\_Sum} \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}$ . Thus, we have

$$\begin{aligned} \Pr[F] &= \Pr[\exists \text{ even } k : G_\tau \text{ contains a path of length } k \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \\ &\leq \sum_{k'=2}^{\infty} \Pr \left[ i_1, \dots, i_{k'+1} : \begin{array}{l} \tau \text{ realizes } (i^{k'+1}, t^{k'}) \\ \bigoplus_{j \in I_1} H_j^1 \oplus \bigoplus_{j \in I_2} S_j = 0^n \\ \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}} \end{array} \right]. \end{aligned}$$

Let  $|I_1 \setminus I_2| = \alpha$  and  $|I_2| = \beta$ . Therefore,  $\alpha + \beta = k' + 1$ . We can choose  $\beta$  many tuples freely in  $q^\beta$  many ways. For each such choice of the tuple, the number of ways we can choose the tuple for  $I_1$  is  $3q^{\alpha-1}$  (using the condition  $\overline{\text{Constant\_Hash\_Sum}}$ ). Therefore, the number of tuples  $i^{k'+1}$  satisfying the condition  $\bigoplus_{j \in I_1} H_j^1 \oplus \bigoplus_{j \in I_2} S_j = 0^n$  is at most  $3q^{k'}$ . For a fixed such tuple, we bound

$$\Pr[\tau \text{ realizes } (i^{k'+1}, t^{k'}) \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}].$$

As  $\text{FHC}$  and  $\text{inner\_bad}$  does not hold, it follows that  $\mathbb{E}_{\tau'}(G_{1,2})$  is a system of linearly independent equations (by following Lemma 17), where  $G_{1,2}$  is a subgraph of  $(i^{k'+1}, t^{k'})$  by removing all its edges of label 3. Thus, we have

$$\Pr[\tau \text{ realizes } (i^{k'+1}, t^{k'}) \wedge \overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{1}{(2^n - 1)_{k'-r}} \cdot \frac{1}{2^{nr}},$$

where  $r$  is the number of label 3 edges in the graph  $(i^{k'+1}, t^{k'})$ . Thus, by assuming  $q \leq 2^{n-2}$  and  $k' \leq 2^{n-1}$ , we have

$$\Pr[F] \leq \sum_{k'=2}^{\infty} \frac{3q^{k'}}{(2^n - 1)_{k'-r}} \cdot \frac{1}{2^{nr}} \leq 3 \sum_{k'=2}^{\infty} \left( \frac{2q}{2^n} \right)^{k'} \leq \frac{24q^2}{2^{2n}}. \quad (20)$$

Note that,

$$\begin{aligned}
\Pr[X_{\text{id}} \in \Theta_b] &= \Pr[\text{Cycle} \vee \text{Component} \vee \text{Zero\_Label\_Sum} \vee \text{EFHC} \vee \text{inner\_bad}] \\
&= \Pr[(\text{Cycle} \vee \text{Component} \vee \text{Zero\_Label\_Sum}) \wedge (\overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})] \\
&\quad + \Pr[\text{EFHC} \vee \text{inner\_bad}] \\
&= \Pr[\text{Cycle} \wedge (\overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})] + \Pr[\text{Component} \wedge (\overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})] \\
&\quad + \Pr[\text{Zero\_Label\_Sum} \wedge (\overline{\text{EFHC}} \wedge \overline{\text{inner\_bad}})] + \Pr[\text{EFHC} \wedge \overline{\text{inner\_bad}}] \\
&\quad + \Pr[\text{inner\_bad}] \\
&\leq \frac{16nq}{2^n} + \frac{32q^2}{2^{2n}} + 2\left(\frac{4q}{2^n}\right)^n + \frac{8q^3\delta}{2^{2n}} + q(\delta + 2^{-n}) + \binom{q}{2}\epsilon \\
&\quad + \binom{q}{2}\delta \cdot 2^{-n},
\end{aligned}$$

where the last inequality follows from Lemma 14, Eqn. (15), Eqn. (18), Eqn. (19), and Eqn. (20), which completes the proof.  $\square$

### 5.1 Good Transcript Analysis

Let  $\tau = (M_1, (S_1, T_1), H_1^1, H_1^2, X_1, Y_1), \dots, (M_q, (S_q, T_q), H_q^1, H_q^2, X_q, Y_q)$  be a fixed arbitrary good transcript. Since  $\tau$  is a good transcript, associated transcript graph  $G_\tau$  does not contain (a) any even length cycle, (b) the maximum size of its component is at most  $2n$ , and (c) it does not contain any even length path such that the label sum of the path is zero. We now show that the probability of realizing  $\tau$  in the real world is as good as the probability of realizing it in the ideal world. In other words, we have the following result:

**Lemma 21.** *Let  $\tau$  be an arbitrary fixed good transcript. Then,*

$$\frac{\Pr[X_{\text{re}} = \tau]}{\Pr[X_{\text{id}} = \tau]} \geq \left(1 - \frac{q^2}{2^{2n}}\right).$$

*Proof.* We compute the ideal interpolation probability and the real interpolation probability for the transcript  $\tau$ . Computing the ideal interpolation probability is straightforward. Note that,  $(S_i, T_i)$ 's are distributed uniformly over  $\{0, 1\}^{2n} \setminus \{(S_1, T_1), \dots, (S_{i-1}, T_{i-1})\}$ , the hash key  $K$  is sampled uniformly at random and finally the permutation  $P$  has been sampled uniformly at random from the set of all  $n$ -bit permutations. Let  $d$  be the number of distinct values of  $H_i^1$  and hence that of  $H_i^2 \oplus X_i$ . Thus, we have

$$\begin{aligned}
\Pr[X_{\text{id}} = \tau] &= \Pr[\widetilde{\text{RP}}(\text{Tw}_i, (L_i, R_i)) = (S_i, T_i), \forall i \in [q]] \cdot \Pr[P(H_i^1) = X_i \oplus H_i^2, \forall i \in [q]] \\
&\quad \cdot \Pr[H_K(\text{Tw}_i, (L_i, R_i)) = (H_i^1, H_i^2) \forall i \in [q]] \\
&= \prod_{i=1}^{\mu} \frac{1}{(2^{2n})_{q_i}} \cdot \frac{1}{|\mathcal{K}|} \cdot \frac{1}{(2^n)_d},
\end{aligned} \tag{21}$$

where  $\mu$  is the distinct number of tweaks and  $q_i$  is the multiplicity of the  $i$ -th tweak. Naturally,  $q_1 + \dots + q_\mu = q$ . On the other hand, to compute a lower bound

on the real interpolation probability, we need to compute the lower bound on the probability that the following system of equations holds, where  $P_0$  and  $P_1$  are two independent  $n$ -bit permutations:

$$\mathbb{E} = \begin{cases} P_0(X'_1) \oplus P_1^{-1}(X_1 \oplus T_1) = H_1^1 \\ P_0(T'_1) \oplus P_1^{-1}(X_1 \oplus T_1) = S_1 \\ P_0(X'_2) \oplus P_1^{-1}(X_2 \oplus T_2) = H_2^1 \\ P_0(T'_2) \oplus P_1^{-1}(X_2 \oplus T_2) = S_2 \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ P_0(X'_q) \oplus P_1^{-1}(X_q \oplus T_q) = H_q^1 \\ P_0(T'_q) \oplus P_1^{-1}(X_q \oplus T_q) = S_q \end{cases}$$

Since the transcript  $\tau$  is good, the graph  $G_\tau$  is acyclic, its component size is bounded by  $2n$ , and for any even-length path, the label sum of the path is non-zero. Therefore, to compute the real interpolation probability, we have

$$\begin{aligned} \Pr[X_{\text{re}} = \tau] &= \Pr[H_K(M_i) = (H_i^1, H_i^2) \forall i \in [q]] \times \Pr[P(H_i^1) = X_i \oplus H_i^2, \forall i \in [q]] \\ &\quad \times \Pr[\mathbb{E} \text{ holds}] \\ &= \frac{1}{|\mathcal{K}|} \cdot \frac{1}{(2^n)_d} \cdot \frac{1}{2^{2nq}}, \end{aligned} \quad (22)$$

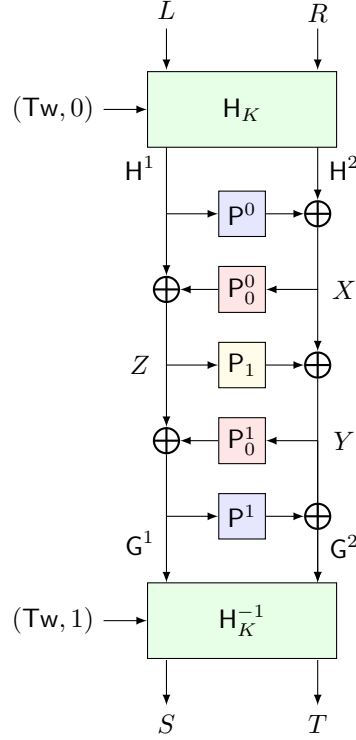
where the last inequality follows from Lemma 2 as  $4n^3 + 2n \leq 2^{n/2}$  for  $n \geq 35$  and  $qn^2 \leq 2^n/12$ . Therefore, by taking the ratio of Eqn. (22) to Eqn. (21), we have

$$\frac{\Pr[X_{\text{re}} = \tau]}{\Pr[X_{\text{id}} = \tau]} = \prod_{i=1}^{\mu} \frac{(2^{2n})_{q_i}}{2^{2nq_i}} \geq \prod_{i=1}^{\mu} \prod_{j=0}^{q_i-1} \frac{(2^{2n} - j)}{2^{2n}} = \prod_{i=1}^{\mu} \prod_{j=0}^{q_i-1} \left(1 - \frac{j}{2^{2n}}\right) \geq \left(1 - \frac{q^2}{2^{2n}}\right),$$

and thus the result follows.  $\square$

## 6 Hash-then-Five round Feistel-then-Hash: An Optimally Secure Tweakable SPRP

In this section, we propose an optimally secure variable input length tweakable SPRP, called  $\text{HF}^5\text{H}$ , which is a composition of an almost universal  $2n$ -bit hash function with a permutation-based five-round Feistel followed by the inverse of the same hash function. It takes  $2n$ -bit input with an arbitrary length tweak, which is fed through a  $2n$ -bit keyed hash function. Let the output be  $(H^1, H^2)$ , which becomes the input of a five-round permutation-based Feistel construction, where all the permutations have been assumed to be sampled independently from the set of all  $n$ -bit permutations. Finally, the output of the five-round Feistel, denoted by  $(G^1, G^2)$ , along with the same tweak, is fed through the inverse of the same hash function and generates the output. The construction is described



**Figure 6.**  $\text{HF}^5\text{H}$ : Hash-then-Five round Feistel-then-Hash Construction.  $\text{H}$  is an  $2n$ -bit almost universal keyed hash function,  $\text{P}$ ,  $\text{P}_0$  and  $\text{P}_1$  are three independent  $n$ -bit permutations. We define the notation  $\text{P}^b(x) := \text{P}(\lfloor x \rfloor \| b)$ .

in Fig. 6 and Algorithm 4. The corresponding decryption algorithm is presented in Algorithm 5.

| Algorithm 4 $\text{HF}^5\text{H}$ Encryption.                                  | Algorithm 5 $\text{HF}^5\text{H}$ Decryption.                                  |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1: <b>function</b> $\text{HF}^5\text{H}.\text{Enc}(K, \text{Tw}, (L, R))$      | 1: <b>function</b> $\text{HF}^5\text{H}.\text{Dec}(K, \text{Tw}, (S, T))$      |
| 2: $(\text{H}^1, \text{H}^2) \leftarrow \text{H}_K(\text{Tw}, 0, (L, R))$      | 2: $(\text{G}^1, \text{G}^2) \leftarrow \text{H}_K(\text{Tw}, 1, (S, T))$      |
| 3: $X \leftarrow \text{P}(\lfloor \text{H}^1 \rfloor \  0) \oplus \text{H}^2$  | 3: $Y \leftarrow \text{P}(\lfloor \text{G}^1 \rfloor \  1) \oplus \text{G}^2$  |
| 4: $Z \leftarrow \text{P}_0(\lfloor X \rfloor \  0) \oplus \text{H}^1$         | 4: $Z \leftarrow \text{P}_0(\lfloor Y \rfloor \  1) \oplus \text{G}^1$         |
| 5: $Y \leftarrow \text{P}_1(Z) \oplus X$                                       | 5: $X \leftarrow \text{P}_1(Z) \oplus Y$                                       |
| 6: $\text{G}^1 \leftarrow \text{P}_0(\lfloor Y \rfloor \  1) \oplus Z$         | 6: $\text{H}^1 \leftarrow \text{P}_0(\lfloor X \rfloor \  0) \oplus Z$         |
| 7: $\text{G}^2 \leftarrow \text{P}(\lfloor \text{G}^1 \rfloor \  1) \oplus Y$  | 7: $\text{H}^2 \leftarrow \text{P}(\lfloor \text{H}^1 \rfloor \  0) \oplus X$  |
| 8: $(S, T) \leftarrow \text{H}_K^{-1}(\text{Tw}, 1, (\text{G}^1, \text{G}^2))$ | 8: $(L, R) \leftarrow \text{H}_K^{-1}(\text{Tw}, 0, (\text{H}^1, \text{H}^2))$ |
| 9: <b>return</b> $(S, T)$                                                      | 9: <b>return</b> $(L, R)$                                                      |

In the following, we show that  $\text{HF}^5\text{H}$  achieves optimal STPRP security. Formally, we have the following result.

**Theorem 4.** *Let  $H$  be an  $(\epsilon, \delta)$ - $2n$ -bit partial zero-sum-universal tweakable keyed hash function. Let  $P, P_0$ , and  $P_1$  be independently sampled  $n$ -bit permutations. Then, for any information theoretic deterministic distinguisher  $A$  that makes  $q$  queries to  $HF^5H[H, P, P_0, P_1]$ , one has*

$$\text{Adv}_{HF^5H[H, P, P_0, P_1]}^{\text{stprp}}(A) \leq \left(\frac{2q}{2^n}\right)^n + q \cdot \left(\frac{4n}{2^n} + \delta\right) + q^2 \cdot \left(\frac{9}{2^{2n}} + \epsilon + \frac{5\delta}{2^n}\right).$$

### 6.1 Proof of Theorem 4

Let  $A$  be a computationally unbounded deterministic distinguisher that interacts with the oracle  $HF^5H$  in the real world or with a  $2n$ -bit tweakable random permutation  $\widetilde{RP}$  in the ideal world. On  $i$ -th query adversary makes either a (i) forward query  $(Tw_i, (L_i, R_i)) \in \mathcal{T} \times \{0, 1\}^{2n}$  and obtain responses  $(S_i, T_i) \in \{0, 1\}^{2n}$  or a (ii) backward query  $(Tw_i, (S_i, T_i)) \in \mathcal{T} \times \{0, 1\}^{2n}$  and obtain responses  $(L_i, R_i) \in \{0, 1\}^{2n}$ . Here,  $\mathcal{T}$  is the tweak space and  $Tw_i$ 's are the tweaks queried by the distinguisher. Let  $\mathcal{E}$  denote the set of all  $i \in [q]$  where the  $i$ -th query is forward, and let  $\mathcal{D} = [q] \setminus \mathcal{E}$  (the set of all query indices corresponding to backward queries). After the interaction, the oracle releases some additional information to the distinguisher (offline phase).

IDEAL WORLD OFFLINE PHASE. In the ideal world, the oracle samples a dummy hash key  $K$  and an  $n$ -bit random permutation  $P$  independently at random. Then, it computes

$$(H_i^1, H_i^2) \leftarrow H_K(Tw_i, 0, (L_i, R_i)), \quad (G_i^1, G_i^2) \leftarrow H_K(Tw_i, 1, (S_i, T_i)).$$

Based on the random permutation  $P$ , we also compute

$$X_i = P(\lfloor H_i^1 \rfloor \| 0) \oplus H_i^2, \quad Y_i = P(\lfloor G_i^1 \rfloor \| 1) \oplus G_i^2, \quad i \in [q].$$

REAL WORLD OFFLINE PHASE. In the real world, the oracle releases the hash key  $K$  and the output of the  $n$ -bit permutation  $P$  for all the  $q$  queries. Then, it computes  $(H_i^1, H_i^2, G_i^1, G_i^2, X_i, Y_i)$  for all  $i \in [q]$ .

ATTACK AND INNER TRANSCRIPT. In both worlds, the attack transcript (the view of the distinguisher after the offline phase is over) is denoted as

$$\tau = ((Tw_1, (L_1, R_1), (S_1, T_1), H_1^1, H_1^2, G_1^1, G_1^2, X_1, Y_1, \text{type}_1), \dots, \\ (Tw_q, (L_q, R_q), (S_q, T_q), H_q^1, H_q^2, G_q^1, G_q^2, X_q, Y_q, \text{type}_q)),$$

where  $\text{type}_i$  denotes whether the  $i$ -th query is in  $\mathcal{E}$  (encryption) or in  $\mathcal{D}$  (decryption). The randomness of the transcript is induced by  $(K, P, (S_i, T_i)_{i \in \mathcal{E}}, (L_i, R_i)_{i \in \mathcal{D}})$ . For any  $i, j \in [q]$ ,  $b \in \{1, 2, 3\}$ , we define

$$\lambda(i, j, b) = \begin{cases} \lfloor H_i^2 \oplus H_j^2 \rfloor & \text{if } b = 1 \\ \lfloor G_i^2 \oplus G_j^2 \rfloor & \text{if } b = 2 \\ \lfloor H_i^2 \oplus G_i^2 \oplus H_j^2 \oplus G_j^2 \rfloor & \text{if } b = 3 \end{cases}$$



The inner transcript (a part of the attack transcript) is defined as

$$\tau' := \tau'(T^q, K) := ((\text{Tw}_1, (L_1, R_1), (S_1, T_1), H_1^1, H_1^2, G_1^1, G_1^2, \text{type}_1), \dots, (\text{Tw}_q, (L_q, R_q), (S_q, T_q), H_q^1, H_q^2, G_q^1, G_q^2, \text{type}_q))$$

Note that,  $\tau'$  is defined over only the tuple  $L^q, R^q, S^q, T^q$  and the hash key  $K$ .  
ATTACK TRANSCRIPT IN THE REAL WORLD. Now we investigate the relationship among the values present in the attack transcript generated in the real world. We first note that

$$P_0(X'_i) \oplus P_1^{-1}(Y_i \oplus X_i) = H_i^1, \quad P_0(Y'_i) \oplus P_1^{-1}(Y_i \oplus X_i) = G_i^1.$$

Note that  $X'_i = X'_j$  is equivalent in saying that

$$[P([H_i^1]||0)] \oplus [P([H_j^1]||0)] = \lambda(i, j, 1).$$

The equation  $Y'_i = Y'_j$  (resp. the equation  $X_i \oplus Y_i = X_j \oplus Y_j$  (after restricting to the most significant  $n - 1$  bits)) implies that

$$[P([G_i^1]||1)] \oplus [P([G_j^1]||1)] = \lambda(i, j, 2).$$

$$[P([H_i^1]||0)] \oplus [P([H_j^1]||0)] \oplus [P([G_i^1]||1)] \oplus [P([G_j^1]||1)] = \lambda(i, j, 3).$$

SYSTEM OF LINEAR EQUATIONS. The above equations motivate us to consider a system of equations associated with an attack transcript of an ideal world. Let  $\{P_h : h \in \{0, 1\}^n\}$  be a collection of  $(n - 1)$ -bit variables. Let  $G = (V \subseteq [q], E, B)$  be a  $\{1, 2, 3\}$ -labeled graph. For any edge  $e$  between  $i$  and  $j$  with a label  $b = B(e) \in \{1, 2, 3\}$ , we associate an equation  $\mathbb{E}_{\tau'}(i, j, b)$  as follows:

$$\mathbb{E}_{\tau'}(i, j, b) : \begin{cases} P_{[H_i^1]||0} \oplus P_{[H_j^1]||0} = \lambda(i, j, 1) & \text{if } b = 1 \\ P_{[G_i^1]||1} \oplus P_{[G_j^1]||1} = \lambda(i, j, 2) & \text{if } b = 2 \\ P_{[H_i^1]||0} \oplus P_{[H_j^1]||0} \oplus P_{[G_i^1]||1} \oplus P_{[G_j^1]||1} = \lambda(i, j, 3) & \text{if } b = 3 \end{cases}$$

We call this graph a *query graph*. If it is a path or a cycle graph, then we call it *query path* or *query cycle* graph respectively. The collection of all equations over the set of all edges of  $G$  is denoted as  $\mathbb{E}_{\tau'}(G)$ .

**Definition 16.** Let  $i^{k+1}$  be an alternating tuple where each  $i_j \in [q]$ . Let  $G$  be a labeled walk over the sequence  $i^{k+1}$  with a label sequence  $b^k$  (i.e.,  $b_j$  is the label of the edge joined by  $i_j$  and  $i_{j+1}$ ). We also denote this labeled walk as  $(i^{k+1}, b^k)$ . We call the labeled walk alternating if  $b^k$  is an alternating sequence and  $b_k \neq b_1$  whenever  $i_{k+1} = i_1$  (i.e., when the walk is a loop).

Now, the system of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  consists of the equations  $\mathbb{E}_{\tau'}(i_1, i_2, b_1)$ ,  $\mathbb{E}_{\tau'}(i_2, i_3, b_2)$ ,  $\dots$ ,  $\mathbb{E}_{\tau'}(i_k, i_{k+1}, b_k)$ .

**Definition 17.** Let  $G$  be a labeled query graph. We say that  $G$  is realizable by an attack transcript  $\tau$  if for all edge  $\{i, j\}$  in  $G$  with a label  $b$ , we have

$$\begin{cases} X'_i = X'_j & \text{if } b = 1 \\ Y'_i = Y'_j & \text{if } b = 2 \\ X_i \oplus Y_i = X_j \oplus Y_j & \text{if } b = 3 \end{cases}$$

**Lemma 22.** *Let  $G$  be a labeled query graph over a vertex set  $V \subseteq [q]$  with label set  $\{1, 2, 3\}$ . If  $\tau$  realizes a labeled graph  $G = (V, E, B)$  then  $(P_h = P(h), \forall h \in \{0, 1\}^n)$  is a solution for the system  $\mathbb{E}_{\tau'}(G)$ .*

*Proof.* For all label 1 edges of  $G$ , it holds that  $X'_i = X'_j$  is same as

$$P_{\lfloor H_i^1 \rfloor \| 0} \oplus P_{\lfloor H_j^1 \rfloor \| 0} = \lambda(i, j, 1).$$

Similarly, for all label 2 edges of  $G$ , it holds that  $Y'_i = Y'_j$  is same as

$$P_{\lfloor G_i^1 \rfloor \| 1} \oplus P_{\lfloor G_j^1 \rfloor \| 1} = \lambda(i, j, 2).$$

Finally, for all label 3 edges of  $G$ , it holds that  $X_i \oplus Y_i = X_j \oplus Y_j$  is same as

$$P_{\lfloor H_i^1 \rfloor \| 0} \oplus P_{\lfloor G_i^1 \rfloor \| 1} \oplus P_{\lfloor H_j^1 \rfloor \| 0} \oplus P_{\lfloor G_j^1 \rfloor \| 1} = \lambda(i, j, 3).$$

Hence,  $P_{\lfloor H_i^1 \rfloor \| 0} = P(\lfloor H_i^1 \rfloor \| 0)$  and  $P_{\lfloor G_i^1 \rfloor \| 1} = P(\lfloor G_i^1 \rfloor \| 1)$  is a solution for the system of equations  $\mathbb{E}_{\tau'}(G)$ .  $\square$

**FIRST HASH COLLISION.** Let us fix a transcript  $\tau$ . For any  $k \geq 1$ , we define the following two sets based on its inner transcript:

$$\begin{aligned} \mathcal{I}_H^{(k)} &:= \{i^{k+1} \in ([q])_{k+1} \mid H_{i_1}^1, \dots, H_{i_k}^1, G_{i_1}^1, \dots, G_{i_{k+1}}^1 \text{ are distinct, } H_{i_1}^1 = H_{i_{k+1}}^1\}, \\ \mathcal{I}_G^{(k)} &:= \{i^{k+1} \in ([q])_{k+1} \mid H_{i_1}^1, \dots, H_{i_{k+1}}^1, G_{i_1}^1, \dots, G_{i_k}^1 \text{ are distinct, } G_{i_1}^1 = G_{i_{k+1}}^1\}. \end{aligned}$$

We define  $\mathcal{I}_H^{(k)} \cup \mathcal{I}_G^{(k)} = \mathcal{I}^{(k)}$ .

**Definition 18.** *We say that first hash-collision (FHC) event holds if there exists an  $k \geq 1$  such that  $i^{k+1} \in \mathcal{I}^{(k)}$  and the attack transcript  $\tau$  realizes an alternating labeled query path  $(i^{k+1}, b^k := (b_1, \dots, b_k))$ .*

**Definition 19.** *We call an inner transcript  $\tau'$  to be inner\_bad if one of the following conditions is satisfied:*

- **Zero\_Hash\_Sum:** For some  $k \geq 2$ ,

$$\left| \{(i_1, i_2, \dots, i_k) : \sum_{j \in I_1} H_j^1 = \sum_{j \in I_2} G_j^1, I_1, I_2 \subseteq \{i_1, \dots, i_k\}\} \right| \geq (3q)^{k-1};$$

- **Univ:**  $\exists i \neq j : (\lfloor H_i^1 \rfloor = \lfloor H_j^1 \rfloor, H_i^2 = H_j^2)$  or  $(\lfloor G_i^1 \rfloor = \lfloor G_j^1 \rfloor, G_i^2 = G_j^2)$ ;
- **Joint\_Coll:** For some  $i, j$ ,  $H_i^1 = H_j^1, G_i^1 = G_j^1$ ;

Let us define an event **inner\_bad** as the union of the above events, i.e.,

$$\text{inner\_bad} := \left( \text{Zero\_Hash\_Sum} \vee \text{Univ} \vee \text{Joint\_Coll} \right).$$

Note that, if an inner transcript  $\tau'$  is bad, then **inner\_bad** event holds. In the following, we upper bound the probability that the **inner\_bad** event holds in the ideal world distribution, proof of which is deferred to Section 6.2.

**Lemma 23.**

$$\Pr[\text{inner\_bad}] \leq q\delta + q^2 \cdot \left( \epsilon + \frac{\delta}{2^n} \right).$$

**Lemma 24.**

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \frac{4q^2\delta}{2^n}.$$

*Proof.* Here we consider two sub-cases depending on whether  $k = 1$  or not.

Case 1:  $k = 1$ . First, consider the subcase  $b = 1$ . Due to  $\overline{\text{Univ}}$ , we only need to consider the subcase when FHC holds due to  $G_{i_1}^1 = G_{i_2}^1$ . In this case, we have two equations:

$$\begin{aligned} P(\lfloor H_{i_1}^1 \rfloor \| 0) \oplus P(\lfloor H_{i_2}^1 \rfloor \| 0) &= \lfloor H_{i_1}^2 \oplus H_{i_2}^2 \rfloor \\ G_{i_1}^1 \oplus G_{i_2}^1 &= 0, \end{aligned}$$

and the event holds with probability  $q(q-1)\delta/(2^n-1)$ . The analysis for  $b = 2$  is similar. For  $b = 3$ , w.l.o.g assume that the hash collision occurs at  $G^1$ , then we have two equations:

$$\begin{aligned} P(\lfloor H_{i_1}^1 \rfloor \| 0) \oplus P(\lfloor H_{i_2}^1 \rfloor \| 0) &= \lfloor H_{i_1}^2 \oplus H_{i_2}^2 \oplus G_{i_1}^2 \oplus G_{i_2}^2 \rfloor \\ G_{i_1}^1 \oplus G_{i_2}^1 &= 0, \end{aligned}$$

Again, it is easy to see that the event holds with probability  $q(q-1)\delta/(2^n-1)$ .

Case 2:  $k \geq 2$ . In this case, there are at most  $(2q)^{k+1}$  possible choices of  $(i^{k+1}, b^k)$  and for any such tuple, the full collision holds with probability at most  $2^{-(n-1)k}$ . Hence, by using the union bound, we have

$$\Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^n \frac{(2q)^{k+1}}{2^{(n-1)k}} \cdot \delta \leq \frac{4q^2\delta}{2^n},$$

where we use the assumption  $k \leq n < 2^{n-1}$ .  $\square$

**Lemma 25.** Suppose FHC and inner\_bad do not occur, then for any labeled tree  $G$ , the system of equations  $\mathbb{E}_{G,\tau'}$  is linearly independent. Hence,

$$\Pr[\tau \text{ realizes } G \wedge \overline{\text{inner\_bad}} \wedge \overline{\text{FHC}}] \leq \frac{1}{(2^n-1)_k}.$$

where  $k$  is the number of edges in  $G$ .

*Proof.* Suppose that the equations are linearly dependent. This implies that there must exist distinct  $i, j$  from  $G$  such that  $H_i^1 = H_j^1$  or  $G_i^1 = G_j^1$ . We choose an  $(i, j)$  pair from  $G$  that has a minimum-length path between them. If this is an alternate path, it is a contradiction as we have assumed that FHC and inner\_bad have not occurred. So, let us assume  $P$  is the non-alternating path from  $i$  to  $j$ . Due to minimality, none of the hash values in the path collide with any other

vertex in the path. Now, let us consider a reduced path where we skip all the nodes that appear with the same label in both neighbouring edges. By definition, this reduced path has an alternating label and hence, FHC must hold (for this reduced path), which gives a contradiction. Hence, the equations are linearly independent, and the probability bound follows.  $\square$

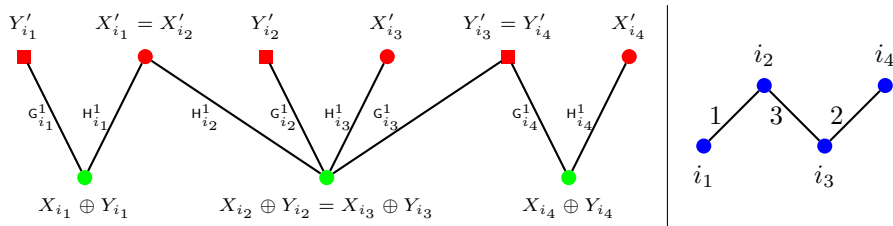
**TRANSCRIPT GRAPH.** We fix an arbitrary transcript  $\tau = (M_1, S_1, T_1, H_1^1, H_1^2, G_1^1, G_1^2, X_1, Y_1, \text{type}_1), \dots, (M_q, S_q, T_q, H_q^1, H_q^2, G_q^1, G_q^2, X_q, Y_q, \text{type}_q)$ . We define a transcript graph  $G_\tau$  corresponding to  $\tau$  as follows:  $G_\tau = (V, \mathcal{E})$  is a bipartite graph that consists of a vertex set  $V_1 \sqcup V_2$  and an edge set  $\mathcal{E}$ .  $V_1$  consists of the variables  $X'_i \cup Y'_i$  for all  $i \in [q]$  and  $V_2$  consists of the variables  $X_i \oplus Y_i$  for all  $i \in [q]$ . For each  $i \in [q]$ , we put a labeled edge between  $X'_i$  and  $X_i \oplus Y_i$  with label  $H_i^1$  and another labeled edge between  $Y'_i$  and  $X_i \oplus Y_i$  with label  $G_i^1$ . If two variable collides (e.g.,  $X'_i = X'_j$  or  $X_i \oplus Y_i = X_j \oplus Y_j$  or  $Y'_i = Y'_j$ ), then we merge the corresponding vertices together.

Recall that, a query graph  $G$  is an edge labeled graph, where the vertices set is the set of query indices and we put an edge between two vertex  $i, j$  with label  $b$  if  $X'_i = X'_j$  (in that case the label is 1) or  $Y'_i = Y'_j$  (in that case the label is 2) or  $X_i \oplus Y_i = X_j \oplus Y_j$  (in that case the label is 3). In the following, we establish a relationship between the transcript graph corresponding to a transcript  $\tau$  and a query graph realizable by  $\tau$ .

**Lemma 26.** *Let  $\tau$  be a fixed transcript and  $G_\tau$  be the corresponding transcript graph.*

1. *If  $G_\tau$  contains an even length cycle, then  $\tau$  realizes an alternating labeled query cycle graph.*
2. *Moreover, if  $G_\tau$  is a tree of size  $2k + 1$  (i.e., the number of vertices in  $G_\tau$ ), then  $\tau$  realizes a query tree  $G$  having size exactly  $k$ .*

*Proof.* Let  $G_\tau$  contain an even-length cycle. Then, by the previous argument, each edge of the cycle gives rise to an equality, which in turn generates an alternating labeled query cycle graph  $G: ((i_1, i_2, \dots, i_k, i_1), b^k)$ . This completes the first part of the proof.



**Figure 7.** A Transcript Graph (left) and its corresponding Query Graph (right)

For the second part of the proof, the number of edges in  $G_\tau$  is  $2k$ . If all the edges in  $G_\tau$  are distinct, then the total number of vertices would have been  $3k$  as two edges give rise to three distinct vertices. But, as the number of vertices in  $G_\tau$  is  $2k + 1$ , there has been a merging of vertices. The number of such merging is  $3k - (2k + 1) = k - 1$ . Note that each merging occurs due to an equality relationship between  $X'$  or  $X \oplus Y$ , or  $Y'$  values. Therefore, the number of such equalities is  $k - 1$ . Each such equality gives rise to an edge in a query graph. Hence, the number of edges in the corresponding query graph realizable by  $\tau$  is  $(k - 1)$ . It is easy to see that, as  $G_\tau$  is a tree, the corresponding query graph realizable by  $\tau$  is also a tree; thus, the number of vertices in the corresponding query graph is  $k$ .  $\square$

**Definition 20 (Bad Attack Transcript).** We call an attainable attack transcript  $\tau$  to be **bad** if either its inner transcript  $\tau'$  is bad or a first hash collision holds or one of the following conditions holds (also called mirror theory bad events):

- (Cycle): if  $G_\tau$  contains an even length cycle.
- (Component): if  $G_\tau$  contains a component with size at least  $2n + 1$ .
- (Zero\_Label\_Sum): if there exists an even length path in  $G_\tau$  such that the label sum of the path is zero.

Let  $X_{\text{id}}$  be the random variable of the transcript realized in the ideal world. Let  $\Theta_b$  denote the set of all bad transcripts. Then, the following lemma upper bounds the probability that a transcript is bad in the ideal world as follows:

**Lemma 27.** Let  $X_{\text{id}}$  and  $\Theta_b$  be defined as above. Then, we have

$$\Pr[X_{\text{id}} \in \Theta_b] \leq 3 \cdot \left(\frac{2q}{2^n}\right)^n + q \cdot \left(\frac{4n}{2^n} + \delta\right) + q^2 \cdot \left(\frac{8}{2^{2n}} + \epsilon + \frac{5\delta}{2^n}\right).$$

The proof of this lemma is deferred to Section 6.3.

**Good Transcript Analysis:** We fix an arbitrary good transcript  $\tau = (M_1, S_1, T_1, H_1^1, H_1^2, G_1^1, G_1^2, X_1, Y_1, \text{type}_1), \dots, (M_q, S_q, T_q, H_q^1, H_q^2, G_q^1, G_q^2, X_q, Y_q, \text{type}_q)$ . Since  $\tau$  is a good transcript, associated transcript graph  $G_\tau$  does not contain (a) any even length cycle, (b) the maximum size of its component is at most  $2n$ , and (c) it does not contain any even length path such that the label sum of the path is zero. We now show that the probability of realizing  $\tau$  in the real world is as good as the probability of realizing it in the ideal world. In other words, we have the following result:

**Lemma 28.** Let  $\tau$  be an arbitrary fixed good transcript. Then,

$$\frac{\Pr[X_{\text{re}} = \tau]}{\Pr[X_{\text{id}} = \tau]} \geq \left(1 - \frac{q^2}{2^{2n}}\right).$$

*Proof.* We compute the ideal interpolation probability and the real interpolation probability for the transcript  $\tau$ . Computing the ideal interpolation probability is straightforward. Note that, for all  $i \in \mathcal{E}$ ,  $(S_i, T_i)$ 's are distributed uniformly over  $\{0, 1\}^{2n} \setminus \{(S_{i_j}, T_{i_j}) : i_j \in \mathcal{E}, i \geq i_j\}$ . Similarly, for all  $i \in \mathcal{D}$ ,  $(L_i, R_i)$ 's are distributed uniformly over  $\{0, 1\}^{2n} \setminus \{(L_{i_j}, R_{i_j}) : i_j \in \mathcal{D}, i \geq i_j\}$ . Moreover, the hash key  $K$  is sampled uniformly at random, and finally the permutation  $P$  has been sampled uniformly at random from the set of all  $n$ -bit permutations. Let  $d_1$  be the number of distinct values of  $\lfloor H_i^1 \rfloor_{n-1}$  and hence that of  $H_i^2 \oplus X_i$ . Suppose  $d_2$  be the number of distinct values of  $\lfloor G_i^1 \rfloor_{n-1}$  and hence that of  $G_i^2 \oplus Y_i$ . Thus, we have

$$\begin{aligned}
\Pr[X_{\text{id}} = \tau] &= \Pr[\widetilde{\text{RP}}(M_i) = (S_i, T_i), \forall i \in \mathcal{E}] \times \Pr[\widetilde{\text{RP}}^{-1}(\text{Tw}_i, S_i, T_i) = (L_i, R_i), \forall i \in \mathcal{D}] \\
&\quad \times \Pr[H_K(\text{Tw}_i, 0, L_i, R_i) = (H_i^1, H_i^2), H_K(\text{Tw}_i, 1, S_i, T_i) = (G_i^1, G_i^2), \forall i \in [q]] \\
&\quad \times \Pr[P(\lfloor H_i^1 \rfloor_{n-1} \| 0) = X_i \oplus H_i^2, \forall i \in [q]] \\
&\quad \times \Pr[P(\lfloor G_i^1 \rfloor_{n-1} \| 1) = Y_i \oplus G_i^2, \forall i \in [q]] \\
&= \prod_{i=1}^{\mu_e} \frac{1}{(2^{2n})_{q_{e,i}}} \times \prod_{i=1}^{\mu_d} \frac{1}{(2^{2n})_{q_{d,i}}} \times \frac{1}{|\mathcal{K}|} \times \frac{1}{(2^n)_{d_1+d_2}}, \tag{23}
\end{aligned}$$

where  $\mu_e$  (resp.  $\mu_d$ ) denotes the distinct number of tweaks in the encryption (resp. decryption) queries. Moreover,  $q_{e,i}$  (resp.  $q_{d,i}$ ) denotes the multiplicity of the  $i$ -th tweak in the encryption (resp. decryption) queries. Note that,  $q_{e,1} + \dots + q_{e,\mu_e} = q_e$ , the total number of encryption queries, and  $q_{d,1} + \dots + q_{d,\mu_d} = q_d$ , the total number of decryption queries. On the other hand, to compute a lower bound on the real interpolation probability, we need to compute the lower bound on the probability that the following system of equations hold, where  $P_0$  and  $P_1$  are two independent  $n$ -bit permutations:

$$\mathbb{E} = \begin{cases} P_0(X'_1) \oplus P_1^{-1}(X_1 \oplus Y_1) = H_1^1 \\ P_0(Y'_1) \oplus P_1^{-1}(X_1 \oplus Y_1) = G_1^1 \\ P_0(X'_2) \oplus P_1^{-1}(X_2 \oplus Y_2) = H_2^1 \\ P_0(Y'_2) \oplus P_1^{-1}(X_2 \oplus Y_2) = G_2^1 \\ \vdots \\ P_0(X'_q) \oplus P_1^{-1}(X_q \oplus Y_q) = H_q^1 \\ P_0(Y'_q) \oplus P_1^{-1}(X_q \oplus Y_q) = G_q^1 \end{cases}$$

Recall that the transcript graph  $G_\tau$ , corresponding to the transcript  $\tau$ , is an edge-labeled bipartite graph. Since the transcript  $\tau$  is good, the graph  $G_\tau$  is acyclic, its component size is bounded by  $2n$ , and for any even-length path, the label sum of the path is non-zero. Therefore, to compute the real interpolation

probability, we have

$$\begin{aligned}
\Pr[X_{\text{re}} = \tau] &= \Pr[H_K(\text{Tw}_i, 0, L_i, R_i) = (H_i^1, H_i^2), \forall i \in [q]] \\
&\quad \times \Pr[H_K(\text{Tw}_i, 1, S_i, T_i) = (G_i^1, G_i^2), \forall i \in [q]] \\
&\quad \times \Pr[P(\lfloor H_i^1 \rfloor_{n-1} \| 0) = X_i \oplus H_i^2, \forall i \in [q]] \\
&\quad \times \Pr[P(\lfloor G_i^1 \rfloor_{n-1} \| 1) = Y_i \oplus G_i^2, \forall i \in [q]] \\
&\quad \times \Pr[P_1(X_i) \oplus P_2^{-1}(X_i \oplus Y_i) = H_i^1, \\
&\quad \quad P_1(Y_i) \oplus P_2^{-1}(X_i \oplus Y_i) = G_i^1, \forall i \in [q]] \\
&= \frac{1}{|\mathcal{K}|} \times \frac{1}{(2^n)_{d_1+d_2}} \times \frac{1}{2^{2nq}}, \tag{24}
\end{aligned}$$

where the last inequality follows from Theorem 2 as we assume  $n \geq 30$  and  $qn^2 \leq 2^n/12$ . Therefore, by taking the ratio of Eqn. (24) to Eqn. (23), we have

$$\begin{aligned}
\frac{\Pr[X_{\text{re}} = \tau]}{\Pr[X_{\text{id}} = \tau]} &= \prod_{i=1}^{\mu_e} \frac{(2^{2n})_{q_{e,i}}}{2^{2nq_{e,i}}} \prod_{i=1}^{\mu_d} \frac{(2^{2n})_{q_{d,i}}}{2^{2nq_{d,i}}} \geq \prod_{i=1}^{\mu_e} \prod_{j=0}^{q_{e,i}-1} \frac{(2^{2n}-j)}{2^{2n}} \prod_{i=1}^{\mu_d} \prod_{j=0}^{q_{d,i}-1} \frac{(2^{2n}-j)}{2^{2n}} \\
&= \prod_{i=1}^{\mu_e} \prod_{j=0}^{q_{e,i}-1} \left(1 - \frac{j}{2^{2n}}\right) \prod_{i=1}^{\mu_d} \prod_{j=0}^{q_{d,i}-1} \left(1 - \frac{j}{2^{2n}}\right) \geq \left(1 - \frac{q_e^2}{2^{2n}}\right) \left(1 - \frac{q_d^2}{2^{2n}}\right) \\
&\geq \left(1 - \frac{q^2}{2^{2n}}\right),
\end{aligned}$$

where the last inequality follows from  $q = q_e + q_d$  and thus the result follows.  $\square$

## 6.2 Proof of Lemma 23

We individually upper bound the probability of the event, and then we combine them by the virtue of the union bound to yield the result.

**Bounding Zero\_Hash\_Sum:** From the definition, we can have at most  $q^{k-1}$  tuples for which zero hash sum holds with probability one. For any other  $q$ -tuples (called non-trivial), zero hash sum holds with probability at most  $\delta$ . Let ZHS denote the expected number of non-trivial zero hash sum  $k$  tuples. By definition,  $\text{Exp}[ZHS] \leq q^k \delta$ . Hence, by Markov inequality,

$$\Pr[ZHS \geq (2q)^{k-1}] \leq \frac{\text{Exp}[ZHS]}{(2q)^{k-1}} \leq \frac{q \delta}{2^{k-1}}.$$

By summing over all  $k$ , we have

$$\Pr[\text{Zero\_Hash\_Sum}] \leq q \delta.$$

**Bounding Univ:** For a fixed choices of indices  $i \neq j$ , the event  $H_K(M_i) = H_K(M_j)$  holds with probability  $\epsilon$ . By varying over all possible choices of indices  $i \neq j$ , we have

$$\Pr[\text{Univ}] \leq \binom{q}{2} \epsilon.$$

**Bounding Joint\_Coll:** We bound the probability of the event as follows. Without loss of generality, assume  $i > j$ .

$$\begin{aligned}
\Pr[\text{Joint\_Coll}] &= \binom{q}{2} \Pr[H_i^1 = H_j^1, G_i^1 = G_j^1] \\
&\leq \binom{q}{2} \Pr[H_K^1(\text{Tw}_i, 0, (L_i, R_i)) = H_K^1(\text{Tw}_j, 0, (L_j, R_j)), \\
&\quad H_K^1(\text{Tw}_i, 1, (S_i, T_i)) = H_K^1(\text{Tw}_j, 1, (S_j, T_j))] \\
&\leq \binom{q}{2} \cdot \delta \cdot 2^{-n}
\end{aligned}$$

The last inequality follows from the randomness of (i)  $K$  and (ii).  $(S_i, T_i)$  (when  $j \in \mathcal{E}$ ) or  $(S_i, T_i)$  (when  $j \in \mathcal{D}$ ).

The lemma follows from applying the union bound to all the above bad events.

### 6.3 Proof of Lemma 27

We have already obtained bounds for the probabilities that an inner transcript is bad. Now, we bound the probability of the subsequent bad events (mirror theory bad events) along with the event **FHC** and **inner\_bad** have not occurred. As **FHC** and **inner\_bad** do not occur, whenever  $\mathbb{E}_k(i^{k+1}, t^k)$  holds (which would appear in the mirror theory bad events), the equations are linearly independent. We use this to bound all remaining bad events.

**A. Bounding  $(\text{Cycle} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contain an even-length cycle. Let  $G$  be the corresponding alternating labeled query graph  $G$  realized from  $\tau$ . To bound the probability of the event that there exists some even  $k$  such that  $G_\tau$  contains a cycle of length  $k$ , we will bound the probability of the event that  $\tau$  realizes an alternating labeled query cycle graph  $G$ . In other words,

Suppose  $\tau$  realizes an alternating labeled query cycle graph  $(i^{k+1}, t^k)$  with  $i_{k+1} = i_1, k \geq 2$ . It is easy to show that the corresponding set of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  is linearly independent. Thus, the number of tuples  $i^k$  satisfying this is at most  $q^k$ . As **FHC** does not hold, we apply Lemma 25 and obtain

$$\Pr[\text{Cycle} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q \frac{n(2q)^{k-1}}{2^{n(k-1)}} \leq \frac{4nq}{2^n}. \quad (25)$$

**B. Bounding  $(\text{Component} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}})$ :** Let  $G_\tau$  contains a component of size exactly  $(2n+1)$ . For the analysis, we concentrate on the largest component of  $G_\tau$  only. Let the component be  $C$ , which is a tree of size  $2n+1$ . Therefore,  $\tau$  realizes a labeled query tree  $G$  consisting of  $n$  vertices. Thus, it contains  $n-1$  edges. Therefore, to bound the probability of the event that  $G_\tau$  contains a component of size at least  $2n+1$ , it is sufficient to bound the probability of the event that  $\tau$  realizes a labeled query tree  $G$  of size  $n$ . Since we consider the event  $\text{FHC} \wedge \text{inner\_bad}$ , following Lemma 25, all the equations in the associated system



of equations  $\mathbb{E}_{\tau'}(G)$  are linearly independent, and by varying the choices for all possible indices, we have

$$\Pr[\text{Component} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \frac{q^{n+1}}{2^{n^2}} \leq \left(\frac{2q}{2^n}\right)^n. \quad (26)$$

**C. Bounding Zero\_Label\_Sum  $\wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}$ :** Let  $G_\tau$  contains an even length path  $P$  of length  $k$  (i.e., the number of edges in  $P$ ),  $k$  is even, such that the label sum of the path  $P$  is zero. Following Lemma 19,  $\tau$  realizes an alternating labeled query path  $P'$ , and the condition that the label sum of the path  $P$  is zero holds. This essentially says that  $\tau$  realizes a set of equations  $\mathbb{E}_{\tau'}(i^{k+1}, b^k)$  and the zero label sum condition holds:

$$\bigoplus_{j \in I_1} H_j^1 \oplus \bigoplus_{j \in I_2} G_j^1 = 0^n,$$

for a fixed  $I_1, I_2 \subseteq \{i_1, \dots, i_{k+1}\}$  derived from  $b^k$ . The number of such tuples  $i^{k+1}$  satisfying the above condition is at most  $(2q)^k$  (using the zero hash sum condition). As FHC does not hold, we apply Lemma 25 and obtain

$$\Pr[\tau \text{ realizes } \mathbb{E}_{\tau'}(i^{k+1}, b^k) \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq 1/(2^n - 1)_k \leq 2/2^{nk}$$

for  $k \leq 2^{n-1}$ . Thus, for  $q \leq 2^{n-2}$ , using the union bound, we have

$$\Pr[\text{Zero_Label_Sum} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \leq \sum_{k=2}^q 2 \cdot \left(\frac{2q}{2^n}\right)^k \leq \frac{8q^2}{2^{2n}}. \quad (27)$$

Note that,

$$\begin{aligned} \Pr[X_{\text{id}} \in \Theta_b] &= \Pr[\text{inner\_bad} \vee \text{FHC} \vee \text{Cycle} \vee \text{Component} \vee \text{Zero_Label_Sum}] \\ &\leq \Pr[\text{inner\_bad}] + \Pr[\text{FHC} \wedge \overline{\text{inner\_bad}}] + \Pr[\text{Cycle} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \\ &\quad + \Pr[\text{Component} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \\ &\quad + \Pr[\text{Zero_Label_Sum} \wedge \overline{\text{FHC}} \wedge \overline{\text{inner\_bad}}] \\ &\leq \left(\frac{2q}{2^n}\right)^n + q \cdot \left(\frac{4n}{2^n} + \delta\right) + q^2 \cdot \left(\frac{8}{2^{2n}} + \epsilon + \frac{5\delta}{2^n}\right), \end{aligned}$$

where the last inequality follows from Lemma 23, Lemma 24, Eqn. (25), Eqn. (26), and Eqn. (27).  $\square$

## References

1. Aoki, K., Ichikawa, T., Kanda, M., Matsui, M., Moriai, S., Nakajima, J., Tokita, T.: Camellia: A 128-bit block cipher suitable for multiple platforms - design and analysis. In: Stinson, D.R., Tavares, S.E. (eds.) Selected Areas in Cryptography, 7th Annual International Workshop, SAC 2000, Waterloo, Ontario, Canada, August 14-15, 2000, Proceedings. Lecture Notes in Computer Science, vol. 2012, pp. 39–56. Springer (2000). [https://doi.org/10.1007/3-540-44983-3\\_4](https://doi.org/10.1007/3-540-44983-3_4), [https://doi.org/10.1007/3-540-44983-3\\_4](https://doi.org/10.1007/3-540-44983-3_4)

2. Bellare, M., Krovetz, T., Rogaway, P.: Luby-Rackoff backwards: Increasing security by making block ciphers non-invertible. pp. 266–280 (1998). <https://doi.org/10.1007/BFb0054132>
3. Bellare, M., Rogaway, P.: The security of triple encryption and a framework for code-based game-playing proofs. In: Vaudenay, S. (ed.) *Advances in Cryptology - EUROCRYPT 2006*, 25th Annual International Conference on the Theory and Applications of Cryptographic Techniques, St. Petersburg, Russia, May 28 - June 1, 2006, Proceedings. Lecture Notes in Computer Science, vol. 4004, pp. 409–426. Springer (2006). [https://doi.org/10.1007/11761679\\_25](https://doi.org/10.1007/11761679_25), [https://doi.org/10.1007/11761679\\_25](https://doi.org/10.1007/11761679_25)
4. Chakraborty, B., Saha, A.: Tweakable permutation-based luby-rackoff constructions. *Cryptology ePrint Archive*, Paper 2025/914 (2025), <https://eprint.iacr.org/2025/914>
5. Cogliati, B., Dutta, A., Nandi, M., Patarin, J., Saha, A.: Proof of mirror theory for a wide range of  $\xi_{\max}$ . pp. 470–501 (2023). [https://doi.org/10.1007/978-3-031-30634-1\\_16](https://doi.org/10.1007/978-3-031-30634-1_16)
6. Cogliati, B., Jha, A., Nandi, M.: How to build optimally secure PRFs using block ciphers. pp. 754–784 (2020). [https://doi.org/10.1007/978-3-030-64837-4\\_25](https://doi.org/10.1007/978-3-030-64837-4_25)
7. Cogliati, B., Seurin, Y.: EWCDM: An efficient, beyond-birthday secure, nonce-misuse resistant MAC. pp. 121–149 (2016). [https://doi.org/10.1007/978-3-662-53018-4\\_5](https://doi.org/10.1007/978-3-662-53018-4_5)
8. Dai, W., Hoang, V.T., Tessaro, S.: Information-theoretic indistinguishability via the chi-squared method. pp. 497–523 (2017). [https://doi.org/10.1007/978-3-319-63697-9\\_17](https://doi.org/10.1007/978-3-319-63697-9_17)
9. Datta, N., Dutta, A., Nandi, M., Paul, G., Zhang, L.: Single key variant of PMAC\_Plus **2017**(4), 268–305 (2017). <https://doi.org/10.13154/tosc.v2017.i4.268-305>
10. Dolmatov, V.: GOST 28147-89: Encryption, Decryption, and Message Authentication Code (MAC) Algorithms. RFC 5830 (Mar 2010). <https://doi.org/10.17487/RFC5830>, <https://www.rfc-editor.org/info/rfc5830>
11. Guo, C., Zhang, G.: Beyond-birthday security for permutation-based feistel networks. *Designs, Codes and Cryptography* **89**, 407–440 (2021)
12. Knudsen, L.: Deal-a 128-bit block cipher. complexity **258**(2), 216 (1998)
13. Luby, M., Rackoff, C.: Pseudo-random permutation generators and cryptographic composition. pp. 356–363 (1986). <https://doi.org/10.1145/12130.12167>
14. Mennink, B., Neves, S.: Encrypted Davies-Meyer and its dual: Towards optimal security using mirror theory. pp. 556–583 (2017). [https://doi.org/10.1007/978-3-319-63697-9\\_19](https://doi.org/10.1007/978-3-319-63697-9_19)
15. Nachev, V., Patarin, J., Volte, E.: *Feistel ciphers*. Cham: Springer International Publishing (2017)
16. Naito, Y.: Blockcipher-based MACs: Beyond the birthday bound without message length. pp. 446–470 (2017). [https://doi.org/10.1007/978-3-319-70700-6\\_16](https://doi.org/10.1007/978-3-319-70700-6_16)
17. Naor, M., Reingold, O.: On the construction of pseudorandom permutations: Luby-Rackoff revisited **12**(1), 29–66 (Jan 1999). <https://doi.org/10.1007/PL00003817>
18. Patarin, J.: Pseudorandom permutations based on the des scheme. In: *International Symposium on Coding Theory and Applications (held in Europe)*. pp. 193–204. Springer (1990)
19. Patarin, J.: A proof of security in  $o(2n)$  for the xor of two random permutations. In: Safavi-Naini, R. (ed.) *Information Theoretic Security, Third International Conference, ICITS 2008*, Calgary, Canada, August 10-13, 2008, Proceed-

- ings. Lecture Notes in Computer Science, vol. 5155, pp. 232–248. Springer (2008), [https://doi.org/10.1007/978-3-540-85093-9\\_22](https://doi.org/10.1007/978-3-540-85093-9_22)
20. Patarin, J.: Introduction to mirror theory: Analysis of systems of linear equalities and linear non equalities for cryptography. Cryptology ePrint Archive, Paper 2010/287 (2010), <https://eprint.iacr.org/2010/287>
  21. Patarin, J.: Introduction to mirror theory: Analysis of systems of linear equalities and linear non equalities for cryptography. IACR Cryptol. ePrint Arch. p. 287 (2010), <http://eprint.iacr.org/2010/287>
  22. Piret, G.: Luby–rackoff revisited: On the use of permutations as inner functions of a feistel scheme. Designs, Codes and Cryptography **39**, 233–245 (2006)
  23. Shirai, T., Shibutani, K., Akishita, T., Moriai, S., Iwata, T.: The 128-bit block-cipher clefia (extended abstract). In: Biryukov, A. (ed.) Fast Software Encryption. pp. 181–195. Springer Berlin Heidelberg, Berlin, Heidelberg (2007)
  24. Yasuda, K.: The sum of CBC macs is a secure PRF. In: Pieprzyk, J. (ed.) Topics in Cryptology - CT-RSA 2010, The Cryptographers’ Track at the RSA Conference 2010, San Francisco, CA, USA, March 1-5, 2010. Proceedings. Lecture Notes in Computer Science, vol. 5985, pp. 366–381. Springer (2010). [https://doi.org/10.1007/978-3-642-11925-5\\_25](https://doi.org/10.1007/978-3-642-11925-5_25), [https://doi.org/10.1007/978-3-642-11925-5\\_25](https://doi.org/10.1007/978-3-642-11925-5_25)
  25. Yasuda, K.: A new variant of PMAC: Beyond the birthday bound. pp. 596–609 (2011). [https://doi.org/10.1007/978-3-642-22792-9\\_34](https://doi.org/10.1007/978-3-642-22792-9_34)
  26. Zhang, L., Wu, W., Sui, H., Wang, P.: 3kf9: Enhancing 3GPP-MAC beyond the birthday bound. pp. 296–312 (2012). [https://doi.org/10.1007/978-3-642-34961-4\\_19](https://doi.org/10.1007/978-3-642-34961-4_19)